



NCIE-017

Implementation Roadmap & Build Plan

**In development — for human review of full expansion —
Version 1.0**

Converts the approved NCIE-001–016 foundation into a controlled implementation sequence: baseline freeze register, Human Review dependency intake, requirement-to-implementation traceability, Work Breakdown Structure, work-package standard, dependency graph, build sequencing/critical path, parallelization, milestones/stage-gates, entry/exit criteria, architecture-to-implementation handover, backlog, nine implementation workstreams (foundation, security/IAM, data/storage, integration, ARGUS/Memory/Agent/AI, Evidence/audit, functional modules, UX, DevSecOps), testing integration, environment/configuration sequencing, migration sequencing, external-system/provider sequencing, operational-readiness sequencing, release/cutover/go-live dependency planning, rollback/recovery/contingency planning, resource/skill planning, estimate/schedule governance, procurement tracking, RAID register, change control/re-baselining, progress reporting, Definition of Ready/Done, and acceptance-gate integration. This document sequences approved work; it does not execute it, and nothing in it may be cited as evidence that implementation has started, tests have passed, or release/go-live has been authorized.

Planning-only, upstream-consuming: NCIE-017 sequences implementation work, dependencies, gates and handovers across the approved NCIE-001–016 foundation; it does not redesign that foundation, resolve its Human Review decisions, fabricate dates/budgets/resources, or claim implementation/test/acceptance has occurred.

Expands approved NCIE-017 Skeletal v0.1, itself traced to approved NCIE-001 v1.3 through approved NCIE-016 v1.0.

Prepared: 16 September 2026

Maintained by: National Communications Intelligence Ecosystem (NCIE)

Document Governance

0.1 Formal-Title Reconciliation (Directive Item 3)

The approved NCIE-017 Skeletal v0.1 carries the working title 'NCIE-017 — Implementation Roadmap & Work Breakdown Specification' and its own Title-control note states that the supplied NCIE-001–015 chain confirms the mandate but does not include the controlled suite register, directing that the exact formal wording be reconciled against that register before full expansion. The authoritative source located and used is the NCIE Documentation Suite Master Controlled Register v2.0 (FOR APPROVAL, dated 02 September 2026), Section 3, whose NCIE-017 entry records the formal title as 'Implementation Roadmap & Build Plan' with Primary Purpose 'Converts approved specifications into implementation sequence, dependencies, work packages and delivery controls' — matching this document's actual 40-chapter scope (WBS Chapter 6, dependency graph Chapter 8, nine implementation workstreams Chapters 15-23, release/delivery controls Chapters 29-30 and 38) exactly. Comparison result: SEMANTIC MATCH — DOCUMENT-CONTROL WORDING DIFFERS (the register uses 'Build Plan' where the skeleton's working title used 'Work Breakdown Specification'; both describe the same underlying sequencing/delivery-control document, and the register's own Primary Purpose text matches the skeleton's actual chapter content precisely, so this is a wording correction only, per directive item 3, not a material scope discrepancy requiring Human Review halt). Per this result, the document-control formal title has been set to 'NCIE-017 — Implementation Roadmap & Build Plan' throughout this document (cover, this section, Document Governance table, Authority Chain, running header, Chapter 40 disposition statement, core document properties and the deliverable filename); descriptive prose referring to the document's 'work breakdown' content throughout the chapters is unaffected, since that describes subject matter (Chapter 6's WBS architecture specifically), not the formal document-control title. This determination is recorded as Human Review item HR17-1-1 for institutional confirmation rather than treated as silently final.

0.2 Authority Chain

NCIE-001 requirements → NCIE-002 architecture → NCIE-003 canonical data → NCIE-004 engineering baseline → NCIE-005 ARGUS behavioral authority → NCIE-006 Memory & Context architecture → NCIE-007 AI/Agent/Model Orchestration → NCIE-008 Governance, Authority, Accountability & RACI → NCIE-009 Security, IAM & Zero-Trust Architecture → NCIE-010 Evidence, Provenance & Audit Architecture → NCIE-011 API, Integration & Data Connector Specification → NCIE-012 UX/UI & Design System Specification → NCIE-013 Functional Modules Specification → NCIE-014 Database & Storage Engineering Specification → NCIE-015 DevSecOps, Infrastructure & Deployment Specification → NCIE-016 Testing, Verification & Acceptance Specification → NCIE-017 Implementation Roadmap & Build Plan. NCIE-017 sequences the approved architecture, security, data, domain, persistence, operational and verification decisions of NCIE-001–016 into a controlled build order; it creates no canonical meaning, institutional authority, security policy, Evidence semantics, API contract, UX pattern, domain-business rule, persistence semantic, operational procedure or verification result of its own — and, as the final document in this suite, its sole substantive basis is the already-approved sixteen specifications that precede it.

0.3 Token-Conservation Convention

Every chapter opens with a short Upstream block citing the specific upstream NCIE-001–016 authority the chapter sequences, stating only the planning/sequencing relationship rather than restating upstream architectural, security, canonical, domain-business, persistence, operational or verification content.

0.4 Governing Implementation-Roadmap Invariants Register

These 19 invariants bind every chapter of this document and are stated once here rather than re-derived repeatedly. They are fixed by the approved NCIE-017 Skeletal v0.1 and by the approved NCIE-001–016 authority chain; no institutional decision is required to adopt them. The upstream canonical, governance, security, Evidence/provenance, integration, UX, domain-module, persistence, operational and verification invariants already binding from NCIE-003 and NCIE-008–016 remain fully in force and are cross-referenced, not restated, in the chapters below.

- ROADMAP ≠ IMPLEMENTATION.
- WORK PACKAGE DEFINED ≠ WORK STARTED.

- WORK STARTED ≠ WORK COMPLETED.
- DEPENDENCY IDENTIFIED ≠ DEPENDENCY RESOLVED.
- SEQUENCED ≠ SCHEDULED DATE.
- ESTIMATE ≠ COMMITMENT.
- BLOCKER RECORDED ≠ BLOCKER CLEARED.
- HUMAN REVIEW ITEM ≠ IMPLEMENTATION DECISION.
- ARCHITECTURE APPROVED ≠ BUILD AUTHORIZED.
- BUILD COMPLETE ≠ TEST PASSED.
- TEST PASSED ≠ PRODUCTION ACCEPTANCE.
- CODE MERGED ≠ RELEASE AUTHORIZED.
- DEPLOYED ≠ APPROVED.
- HANDOVER PACKAGE ≠ CREDENTIAL TRANSFER.
- PARALLELIZABLE ≠ INDEPENDENT.
- CRITICAL PATH ≠ INSTITUTIONAL PRIORITY.
- MILESTONE REACHED ≠ ACCEPTANCE GATE PASSED.
- AUTOMATION ≠ UNBOUNDED DELIVERY AUTHORITY.
- CHANGE IN SEQUENCE ≠ CHANGE IN ARCHITECTURE.

A twentieth invariant, IMPLEMENTATION EVIDENCE ≠ ACCEPTANCE EVIDENCE UNLESS GOVERNED AS SUCH, binds Chapters 24 and 38 specifically and is restated at point of use.

0.5 Planning/Upstream Boundary Note

NCIE-017 sequences implementation work, dependencies, gates and handovers across the approved NCIE-001–016 foundation. It consumes that foundation as its sole substantive basis; it does not redefine any of it, and it does not invent implementation authority, institutional priority, risk acceptance, or release/go-live authority, and it does not silently resolve Human Review. Where a matter requires a calendar date, duration, budget, named individual, team capacity, procurement lead time, delivery commitment, or resource-availability figure not fixed by an approved source, this document states the sequencing logic and marks the value TBD / Human Review rather than inventing one. Approval of the NCIE-017 Skeletal v0.1 authorized this full expansion only; it did not itself approve any date, budget, resource assignment, platform selection or release authority proposed below — and it did not authorize a single unit of implementation work to begin.

0.6 Human Review Focus Classification

Every chapter's Human Review Focus item carries one of seven labels: RESOLVED BY UPSTREAM BASELINE; PROPOSED DESIGN DEFAULT; INSTITUTIONAL CONFIRMATION REQUIRED; SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED; LEGAL/POLICY CONFIRMATION REQUIRED; ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED; or SOURCE/WORKFLOW DISCOVERY REQUIRED — each additionally marked Blocking or Non-Blocking, and each stating precisely what it blocks. Per directive item 73, a Chapter 39 register item that restates an identical HR1–HR16 question already open upstream cross-references that upstream Human Review ID instead of duplicating it; a new NCIE-017 item is created only for a distinct roadmap/sequencing decision. Chapter 39 consolidates every open item, extracted directly from the chapters so the register cannot drift from its sources (mechanically verified per §39.4).

0.7 Status and No-Execution-Claims Convention

Skeletal-structure approval authorizes expansion; it does not approve any individual date, budget, resource assignment, platform selection or release authority proposed below — those remain IN DEVELOPMENT / FOR HUMAN REVIEW until a separate approval act. Per directive item 81, this document must never claim the roadmap has been executed, resourced, funded, scheduled, tested or accepted merely because it has been specified; no work package in this document has actually started, no test referenced from NCIE-016 has actually been executed, and no release/go-live has actually been authorized. This convention is restated in every chapter's status line so it cannot be overlooked.

0.8 Salient Things to Note Discipline

Directive item 85 requires this expansion to programmatically verify an explicit '<N>.0 Salient Things to Note' subsection in all 40 chapters, continuing the discipline established after the NCIE-009 Chapter-27 omission (found by independent human examination and corrected) and reinforced in every full expansion since. Every chapter below

renders its Salient Things to Note subsection as an unconditional part of the same build-time helper (NCIE-017's lib.py raises a build error if a chapter's salient content is missing), and Chapter 40's mechanical audit (§40.2) re-verifies this against the final document.

0.9 Document Amendment History

Version: v0.1

Change: Approved skeletal review edition (structural/conceptual review only; full expansion not yet authorized).

Version: v1.0

Change: Full expansion of the approved NCIE-017 Skeletal v0.1 into the complete 40-chapter specification per the NCIE-017 v0.1→v1.0 Claude Full-Expansion Production Directive: eight consolidated matrices/registers (Master WBS, Dependency Register, Roadmap Sequence Matrix, Stage-Gate Matrix, Architecture-to-Implementation Handover Matrix, HR1–HR16 Dependency Matrix, Resource/Skill Matrix, RAID Register), schedule/progress-fabrication and architecture-drift audits, Human Review Register (Chapter 39) and full traceability/final handover (Chapter 40). Formal title reconciled against the NCIE Documentation Suite Master Controlled Register v2.0 at production time (§0.1) — document-control wording corrected to 'Implementation Roadmap & Build Plan'; no material scope discrepancy found.

Authority: Approved NCIE-001 Master PRD v1.3 through approved NCIE-016 Testing, Verification & Acceptance Specification v1.0, and the approved NCIE-017 Skeletal v0.1.

Scope: Controlled full expansion only — the approved 40-chapter skeletal structure, chapter purposes and boundaries are preserved exactly; no chapter was added, removed, merged, split or renumbered. This expansion authorizes roadmap/specification production only; it does not authorize implementation execution.

Status: v1.0 IN DEVELOPMENT — FOR HUMAN REVIEW OF FULL EXPANSION. The approved v0.1 skeleton remains historically preserved as the expansion architecture.

Document	NCIE-017 Implementation Roadmap & Build Plan
Edition	In development — Version 1.0 (Full Expansion of Approved Skeletal v0.1)
Requirements authority	NCIE-001 Master Product Requirements Document, Version 1.3 — APPROVED
Architecture authority	NCIE-002 System Architecture & Technical Design, Version 0.5 — APPROVED
Canonical data authority	NCIE-003 Data Model & Data Dictionary, Version 0.4 — APPROVED
Engineering baseline	NCIE-004 Technology Stack & Engineering Blueprint, Version 1.2 — APPROVED
ARGUS behavioral authority	NCIE-005 ARGUS Intelligence Assistant Specification, Version 1.3 — APPROVED
Memory & Context authority	NCIE-006 Memory & Context Architecture Specification, Version 1.2 — APPROVED
Orchestration authority	NCIE-007 AI, Agent & Model Orchestration Specification, Version 1.2 — APPROVED
Governance authority	NCIE-008 Governance, Authority, Accountability & RACI Specification, Version 1.0 — APPROVED
Security authority	NCIE-009 Security, IAM & Zero-Trust Architecture, Version 1.0 — APPROVED
Evidence/provenance/audit authority	NCIE-010 Evidence, Provenance & Audit Architecture, Version 1.0 — APPROVED

Integration authority	NCIE-011 API, Integration & Data Connector Specification, Version 1.0 — APPROVED
UX/UI authority	NCIE-012 UX/UI & Design System Specification, Version 1.0 — APPROVED
Domain-module authority	NCIE-013 Functional Modules Specification, Version 1.0 — APPROVED
Persistence authority	NCIE-014 Database & Storage Engineering Specification, Version 1.0 — APPROVED
Operations authority	NCIE-015 DevSecOps, Infrastructure & Deployment Specification, Version 1.0 — APPROVED
Verification authority	NCIE-016 Testing, Verification & Acceptance Specification, Version 1.0 — APPROVED
Prior baseline	NCIE-017 Implementation Roadmap & Work Breakdown Specification (working title), Skeletal v0.1 — APPROVED FOR FULL EXPANSION (historically preserved)
Status	IN DEVELOPMENT — FOR HUMAN REVIEW OF FULL EXPANSION until explicitly approved; the approved v0.1 skeleton remains historically preserved as the expansion architecture. No implementation work has started under this document.

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Mandate, Authority Chain & Roadmap Boundary

Establish NCIE-017 ownership of implementation sequencing and work breakdown within the approved NCIE suite.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Planning specification only; nothing in this chapter implies implementation work has started, tests have passed, blockers have cleared, or release/go-live is authorized.

Upstream: NCIE-002 Ch.32; NCIE-014 Ch.34; approved NCIE-001–016 chain.

1.0 Salient Things to Note

NCIE-017 sequences approved work; it cannot redesign NCIE-001–016 or claim implementation occurred — Table 1 fixes this boundary so no later chapter can drift into claiming ownership of what an upstream document already owns.

1.1 Mandate and Scope

NCIE-017 sequences approved work across NCIE-001–016 into a controlled implementation roadmap — it cannot redesign that foundation or claim implementation has occurred. Roadmap approval is not execution authorization.

1.2 Roadmap Boundary — What NCIE-017 Does Not Own

NCIE-017 does not own product requirements, architecture, canonical data, ARGUS behavior, Memory/Context semantics, Agent/VPF semantics, institutional governance, security policy, Evidence/provenance meaning, API contracts, UX semantics, domain business rules, persistence semantics, operational procedures or verification results. It owns only the sequencing layer above all of them.

Table 1. NCIE-017 roadmap boundary.

Concern	Owned By
Product requirements/scope	NCIE-001
System architecture	NCIE-002
Canonical data model	NCIE-003
Engineering baseline/technology families	NCIE-004
ARGUS behavioral authority	NCIE-005
Memory & Context architecture	NCIE-006
AI/Agent/Model orchestration	NCIE-007
Institutional governance, RACI	NCIE-008
Security, IAM, Zero-Trust enforcement	NCIE-009
Evidence lifecycle, provenance, audit	NCIE-010
API/event/connector/integration contracts	NCIE-011
UX/UI, presentation, interaction, Design System	NCIE-012
Domain business rules, workflows, analytical products	NCIE-013
Physical database/storage engineering	NCIE-014
Deployment/infrastructure realization, operations	NCIE-015
Verification strategy, RTM, evidence, acceptance	NCIE-016
Implementation sequencing, WBS, roadmap, handover	NCIE-017 (this document)

1.3 Roadmap Outputs and Handovers

NCIE-017's outputs are the Master WBS (Chapter 6), dependency register (Chapter 8), roadmap sequence and stage-gate matrices (Chapters 9, 12), the architecture-to-implementation handover package (Chapter 13), and the consolidated Human Review Register (Chapter 39) — each an output of controlled planning, never of executed work.

1.R RACI / Control Ownership

Table 2. Chapter 1 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Roadmap)	Consulted	Informed
Adopt/confirm NCIE-017 roadmap architecture	Accountable Authority Class (NCIE-008, to be assigned)	Roadmap Architecture Owner (to be assigned)	NCIE-001-016 chapter owners	All institutional role holders

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR17-1-1: Confirm the NCIE-017 formal title reconciliation (\$0.1 — corrected to 'Implementation Roadmap & Build Plan' against Master Controlled Register v2.0), and roadmap acceptance authority (skeletal Preliminary Human Review Theme 1).

Traceability: NCIE-002 Ch.32; NCIE-014 Ch.34; approved NCIE-001–016 chain.

Implementation Planning Doctrine

Define disciplined implementation-planning principles governing every chapter this document specifies.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Planning specification only; nothing in this chapter implies implementation work has started, tests have passed, blockers have cleared, or release/go-live is authorized.

Upstream: NCIE-001–016.

2.0 Salient Things to Note

ROADMAP ≠ IMPLEMENTATION and ESTIMATE ≠ COMMITMENT — sequence, estimate, commitment, execution and acceptance are five distinct acts that this document never collapses into one.

2.1 Sequence, Estimate, Commitment, Execution and Acceptance Are Distinct

ROADMAP ≠ IMPLEMENTATION and ESTIMATE ≠ COMMITMENT — sequencing work logically, estimating its effort, committing to a delivery, executing it, and accepting the result are five distinct acts, and this document repeatedly separates them because collapsing any two is the most common way a roadmap silently overclaims.

2.2 Evidence, Uncertainty, Human Control and Anti-Fabrication

Every planning statement is evidence-based (traced to an approved upstream source or an explicit Human Review item) and carries its uncertainty honestly; Human control over institutional priority, risk acceptance and release authority is preserved throughout, and no date, budget, resource or status is fabricated to make the roadmap look more complete than it is.

2.R RACI / Control Ownership

Table 3. Chapter 2 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Roadmap)	Consulted	Informed
Approve planning doctrine and roadmap ownership	Accountable Authority Class (to be assigned)	Planning Doctrine Owner (to be assigned)	NCIE-008 owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR17-2-1: Approve planning doctrine and roadmap ownership.

Traceability: NCIE-001–016.

Approved Baseline & Architecture Freeze Register

Freeze and reference the Human-approved architecture baseline used for planning.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Planning specification only; nothing in this chapter implies implementation work has started, tests have passed, blockers have cleared, or release/go-live is authorized.

Upstream: NCIE-001–016.

3.0 Salient Things to Note

Only approved/current baselines may drive build sequencing; superseded versions remain historical — Table 4 freezes the exact version of every upstream specification this roadmap plans against.

3.1 Document/Version Register

Only approved/current baselines may drive build sequencing; superseded versions remain historical reference only. The register below freezes the exact version of every upstream specification this roadmap plans against.

Table 4. Approved Baseline & Architecture Freeze Register.

Document	Approved Version	Status	Freeze Point
NCIE-001 Master PRD	v1.3	APPROVED	This roadmap plans against v1.3
NCIE-002 System & Solution Architecture	v0.5	APPROVED	This roadmap plans against v0.5
NCIE-003 Data Model & Data Dictionary	v0.4	APPROVED	This roadmap plans against v0.4
NCIE-004 Technology Stack & Engineering Blueprint	v1.2	APPROVED	This roadmap plans against v1.2
NCIE-005 ARGUS Intelligence Assistant Specification	v1.3	APPROVED	This roadmap plans against v1.3
NCIE-006 Memory & Context Architecture Specification	v1.2	APPROVED	This roadmap plans against v1.2
NCIE-007 AI, Agent & Model Orchestration Specification	v1.2	APPROVED	This roadmap plans against v1.2
NCIE-008 Governance, Authority, Accountability & RACI	v1.0	APPROVED	This roadmap plans against v1.0
NCIE-009 Security, IAM & Zero-Trust Architecture	v1.0	APPROVED	This roadmap plans against v1.0
NCIE-010 Evidence, Provenance & Audit Architecture	v1.0	APPROVED	This roadmap plans against v1.0
NCIE-011 API, Integration & Data Connector Specification	v1.0	APPROVED	This roadmap plans against v1.0
NCIE-012 UX/UI & Design System Specification	v1.0	APPROVED	This roadmap plans against v1.0

NCIE-013 Functional Modules Specification	v1.0	APPROVED	This roadmap plans against v1.0
NCIE-014 Database & Storage Engineering Specification	v1.0	APPROVED	This roadmap plans against v1.0
NCIE-015 DevSecOps, Infrastructure & Deployment Specification	v1.0	APPROVED	This roadmap plans against v1.0
NCIE-016 Testing, Verification & Acceptance Specification	v1.0	APPROVED	This roadmap plans against v1.0

3.2 Supersession and Re-Baselining

If any frozen document above is later amended, this register's freeze point is stale for the affected scope and Chapter 35's change-control process governs re-baselining; this document does not silently re-plan from an unreviewed newer version.

3.R RACI / Control Ownership

Table 5. Chapter 3 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Roadmap)	Consulted	Informed
Confirm baseline-freeze authority/version register	Accountable Authority Class (to be assigned)	Baseline Register Owner (to be assigned)	NCIE-001-016 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR17-3-1: Confirm baseline-freeze authority/version register (Table 4).

Traceability: NCIE-001–016.

Human Review Dependency Intake

Convert unresolved HR1–HR16 items into explicit build-sequence dependencies.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Planning specification only; nothing in this chapter implies implementation work has started, tests have passed, blockers have cleared, or release/go-live is authorized.

Upstream: NCIE-001–016.

4.0 Salient Things to Note

HUMAN REVIEW ITEM ≠ IMPLEMENTATION DECISION; unresolved blockers stay unresolved — Table 6 intakes upstream Blocking items as visible dependencies without resolving or diluting any of them.

4.1 Intake Discipline

HUMAN REVIEW ITEM ≠ IMPLEMENTATION DECISION — an upstream Blocking Human Review item remains unresolved when it is imported here; intake makes it a visible build-sequence dependency, it does not resolve it, close it, or reduce its blocking scope.

4.2 HR1–HR16 Dependency Matrix (directive item 68)

The consolidated matrix below intakes the Blocking Human Review items from each upstream specification's own register as build-sequence dependencies, distinguishing referenced upstream HR items from new HR17 items created only for distinct roadmap decisions.

Table 6. HR1–HR16 Dependency Matrix (representative entries by source).

Upstream Register	Representative Blocking Theme	Owner Authority Reference	Build-Sequence Impact	Status
NCIE-008 HR8	Governance/RACI accountable-authority assignment	NCIE-008 accountable authority	Blocks Chapters 16, 38 acceptance-gate authority binding	Unresolved upstream — referenced, not cloned
NCIE-009 HR9	Security/IAM/Zero-Trust confirmation items	NCIE-009 owner	Blocks Chapter 16 security/IAM workstream sequencing	Unresolved upstream — referenced, not cloned
NCIE-010 HR10	Evidence/provenance/audit confirmation items	NCIE-010 owner	Blocks Chapter 20 Evidence/audit workstream sequencing	Unresolved upstream — referenced, not cloned
NCIE-013 HR13	Domain rule/KPI/threshold confirmation items	NCIE-013 owner	Blocks Chapter 21 functional-modules workstream sequencing	Unresolved upstream — referenced, not cloned
NCIE-014 HR14	Persistence technology/threshold confirmation items	NCIE-014 owner	Blocks Chapter 17 data/storage workstream sequencing	Unresolved upstream — referenced, not cloned
NCIE-015 HR15	Operations technology/threshold confirmation items	NCIE-015 owner	Blocks Chapter 23 DevSecOps workstream sequencing	Unresolved upstream — referenced, not cloned
NCIE-016 HR16	Test tool/threshold/acceptance-	NCIE-016 owner	Blocks Chapter 24 testing-integration	Unresolved upstream —

authority confirmation
items

and Chapter 38
acceptance-gate
sequencing

referenced, not
cloned

4.R RACI / Control Ownership

Table 7. Chapter 4 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Roadmap)	Consulted	Informed
Confirm treatment/ownership of Blocking upstream HR items	Accountable Authority Class (to be assigned)	HR Intake Owner (to be assigned)	NCIE-001-016 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR17-4-1: Confirm treatment/ownership of Blocking upstream HR items (Table 6).

Traceability: NCIE-001–016.

Requirement-to-Implementation Traceability

Trace approved requirements to implementation work without redefining them.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Planning specification only; nothing in this chapter implies implementation work has started, tests have passed, blockers have cleared, or release/go-live is authorized.

Upstream: NCIE-001–016.

5.0 Salient Things to Note

Traceability maps requirements to work; it does not correct requirements — the chain extends NCIE-016's own RTM structure through implementation and acceptance without redefining any requirement.

5.1 Requirement, Source/Version and Work Linkage

Every implementation traceability entry links a requirement (source and version, per Chapter 3's freeze register) to the WBS/work package that will implement it; traceability maps requirements to work, it does not correct requirements.

5.2 Evidence, Test and Acceptance Linkage

The traceability chain continues from work package through implementation evidence (Chapter 36), NCIE-016 test linkage (Chapter 24) and acceptance state (Chapter 38) — the same structure as NCIE-016's own RTM, extended to cover the implementation and acceptance stages NCIE-016 hands downstream.

5.R RACI / Control Ownership

Table 8. Chapter 5 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Roadmap)	Consulted	Informed
Confirm implementation traceability ownership	Accountable Authority Class (to be assigned)	Traceability Owner (to be assigned)	NCIE-016 owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR17-5-1: Confirm implementation traceability ownership.

Traceability: NCIE-001–016.

Work Breakdown Structure (WBS) Architecture

Define hierarchical WBS structure across the NCIE suite.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Planning specification only; nothing in this chapter implies implementation work has started, tests have passed, blockers have cleared, or release/go-live is authorized.

Upstream: NCIE-001–016.

6.0 Salient Things to Note

WBS decomposition must preserve ownership and upstream semantic boundaries — Table 9 maps every workstream to its single owning upstream specification, never spanning two without an explicit cross-reference.

6.1 Programme, Workstream, Epic, Package and Task Concept

The WBS decomposes as Programme → Workstream → Epic → Work Package → Task, with stable identifiers at every level; WBS decomposition must preserve ownership and upstream semantic boundaries — a work package never spans two upstream specifications' ownership without an explicit cross-reference to both.

6.2 Master WBS (directive item 63)

The consolidated Master WBS below covers all approved NCIE-001–016 implementation domains at the workstream level; each workstream expands into work packages in its own chapter (Chapters 15-23), each carrying a stable ID, source authority, objective, inputs, outputs, dependencies, HR blockers, evidence requirement, and Ready/Done criteria per the Chapter 7 standard.

Table 9. Master WBS (workstream level).

WBS ID	Workstream	Source Authority	Objective	Owning Chapter
WBS-15	Foundation / Platform	NCIE-002/004	Shared foundation/platform capabilities	Ch.15
WBS-16	Security, IAM & Governance	NCIE-008/009	Governance/IAM/Zero-Trust/secrets/audit prerequisites	Ch.16
WBS-17	Data, Database & Storage	NCIE-003/014	Canonical data, persistence, migration, backup/DR	Ch.17
WBS-18	API, Integration & Connector	NCIE-011	Gateway, events, schemas, connectors, external integration	Ch.18
WBS-19	ARGUS, Memory, Agent & AI	NCIE-005/006/007	ARGUS, Memory/Context, Agent Factory/runtime, Model/Tool Gateway, VPF	Ch.19
WBS-20	Evidence, Provenance & Audit	NCIE-010	Evidence, custody, provenance, audit, Decision-Time	Ch.20
WBS-21	Functional Modules	NCIE-013	Domain modules (Network, Traffic, Revenue, Mobile Money, SIM, Anti-Fraud, REWS)	Ch.21

WBS-22	UX/UI & Design System	NCIE-012	Design system, shell, ARGUS UX, accessibility	Ch.22
WBS-23	DevSecOps, Infrastructure & Deployment	NCIE-015	CI/CD, environments, IaC, runtime, observability, incident/runbooks	Ch.23
WBS-24	Testing, Verification & Acceptance	NCIE-016	RTM/test-case integration into build sequence	Ch.24

6.R RACI / Control Ownership

Table 10. Chapter 6 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Roadmap)	Consulted	Informed
Approve WBS hierarchy/identifier convention	Accountable Authority Class (to be assigned)	WBS Owner (to be assigned)	NCIE-001-016 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR17-6-1: Approve WBS hierarchy/identifier convention (Table 9).

Traceability: NCIE-001–016.

Work Package Definition Standard

Define minimum content for every implementable work package.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Planning specification only; nothing in this chapter implies implementation work has started, tests have passed, blockers have cleared, or release/go-live is authorized.

Upstream: NCIE-001–016.

7.0 Salient Things to Note

WORK PACKAGE DEFINED ≠ WORK STARTED; each package needs explicit inputs/outputs/dependencies/evidence — definition never implies the package has begun.

7.1 Objective, Inputs, Outputs and Dependencies

WORK PACKAGE DEFINED ≠ WORK STARTED — every work package states its objective, inputs, outputs, dependencies and owner role at definition time; none of this implies the package has begun, only that it is ready to be scheduled once its dependencies clear.

7.2 Assumptions, Risks, Evidence and Definition of Ready/Done

Every work package records its assumptions and risks explicitly, states the evidence its completion will require, and references the Chapter 37 Definition of Ready/Done criteria rather than each package inventing its own ad hoc completion bar.

7.R RACI / Control Ownership

Table 11. Chapter 7 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Roadmap)	Consulted	Informed
Confirm work-package ownership/evidence standard	Accountable Authority Class (to be assigned)	Work Package Standard Owner (to be assigned)	NCIE-001-016 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR17-7-1: Confirm work-package ownership/evidence standard.

Traceability: NCIE-001–016.

Dependency Graph & Predecessor Logic

Model predecessor/successor and blocking dependencies.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Planning specification only; nothing in this chapter implies implementation work has started, tests have passed, blockers have cleared, or release/go-live is authorized.

Upstream: NCIE-001–016.

8.0 Salient Things to Note

DEPENDENCY IDENTIFIED ≠ DEPENDENCY RESOLVED; circular/hidden dependencies must be exposed — Table 12 illustrates the cross-workstream register structure with every blocker state shown Open, none silently marked cleared.

8.1 Dependency Types and Cycle Detection

DEPENDENCY IDENTIFIED ≠ DEPENDENCY RESOLVED — predecessor, successor, blocking, conditional and external dependency types are each modeled explicitly; the graph is scanned for cycles so a circular dependency is exposed rather than silently deadlocking the sequence.

8.2 Dependency Register (directive item 64)

The consolidated register below tracks cross-workstream dependencies with predecessor/successor, type, evidence, blocker state, HR source and escalation path, illustratively — the full register populates as each workstream chapter's work packages are defined.

Table 12. Cross-Workstream Dependency Register (illustrative structure).

Depend ency ID	Predecess or	Successor	Type	Evidence	Blocker State	HR Source	External Depend ency	Escalation
DEP-16-17	WBS-16 (Security/AM)	WBS-17 (Data/Storage)	Blocking — privileged access required before persistence operations	NCIE-009 Ch.28 privileged-access control delivered	Open (upstream HR9 unresolved)	NCIE-009 HR9	No	Accountable Authority Class per Chapter 4
DEP-16-19	WBS-16 (Security/AM)	WBS-19 (ARGUS/Agent/AI)	Blocking — sandbox isolation required before Agent runtime	NCIE-009 Ch.19 sandbox controls delivered	Open (upstream HR9 unresolved)	NCIE-009 HR9	No	Accountable Authority Class per Chapter 4
DEP-17-21	WBS-17 (Data/Storage)	WBS-21 (Functional)	Blocking — canonical	NCIE-014 Ch.4 mapping	Open (upstream)	NCIE-014	No	Accountable Authority Class

	age)	Modules)	al persiste nce require d before domain - module rule evaluati on	delivered	HR14 unresol ved)	HR14		per Chapter 4
DEP-18-27	WBS-18 (API/Integr ation)	External provider onboarding (Ch.27)	Condi tional — external	Provider contract/certif ication evidence	Open (externa l, not yet discove red)	N/A — exter nal	Yes	Procurement/E xternal Owner (Ch.33)
DEP-20-21	WBS-20 (Evidence/ Audit)	WBS-21 (Functional Modules)	Blockin g — Evidenc e interfac e require d before domain Evidenc e-linked outputs	NCIE-010 Ch.27 defensibility evidence delivered	Open (upstre am HR10 unresol ved)	NCIE -010 HR10	No	Accountable Authority Class per Chapter 4
DEP-23-24	WBS-23 (DevSecOp s)	WBS-24 (Testing/Verifi cation)	Blockin g — CI/CD pipeline require d before automa ted verificat ion executi on	NCIE-015 Ch.37 CI/CD integration delivered	Open (upstre am HR15 unresol ved)	NCIE -015 HR15	No	Accountable Authority Class per Chapter 4

8.R RACI / Control Ownership

Table 13. Chapter 8 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Roadmap)	Consulted	Informed
Confirm dependency classification/escalation	Accountable Authority Class (to be assigned)	Dependency Register Owner (to be assigned)	NCIE-001-016 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR17-8-1: Confirm dependency classification/escalation (Table 12).

Traceability: NCIE-001–016.

Build Sequencing & Critical-Path Method

Define sequence logic and critical-path analysis without inventing dates.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Planning specification only; nothing in this chapter implies implementation work has started, tests have passed, blockers have cleared, or release/go-live is authorized.

Upstream: NCIE-001–016.

9.0 Salient Things to Note

SEQUENCED ≠ SCHEDULED DATE; critical path is analytical, not institutional priority — Table 14 states logical predecessors for all ten workstreams with every Calendar Date cell reading Not assigned.

9.1 Sequence Logic and Float Concept

SEQUENCED ≠ SCHEDULED DATE — logical build order (what must precede what) is analyzed independently of any calendar; float (schedule slack) is a property of the dependency graph, not an institutional priority statement.

9.2 Critical Path and Gate Constraints

CRITICAL PATH ≠ INSTITUTIONAL PRIORITY — the critical path is the longest dependency chain through the graph, an analytical property; which workstream the institution actually prioritizes is a separate, Human Review-governed decision that may or may not follow the critical path.

9.3 Roadmap Sequence Matrix (directive item 65)

The consolidated matrix below states the logical sequence for the ten workstreams (foundation, security, data/storage, integration, AI/ARGUS, Evidence/audit, Functional Modules, UX, DevSecOps, verification) without assigning any calendar date.

Table 14. Roadmap Sequence Matrix.

Workstream	Logical Predecessor(s)	Rationale	Calendar Date
Foundation / Platform (WBS-15)	None (first logical workstream)	Shared runtime/identity/observability foundations underlie every other workstream	Not assigned — TBD/Human Review
Security, IAM & Governance (WBS-16)	Foundation / Platform	Privileged access, secrets and Zero-Trust controls are enabling dependencies for every subsequent workstream	Not assigned — TBD/Human Review
Data, Database & Storage (WBS-17)	Foundation / Platform; Security, IAM & Governance	Persistence requires both platform runtime and access control	Not assigned — TBD/Human Review
API, Integration & Connector (WBS-18)	Data, Database & Storage; Security, IAM & Governance	Integration contracts consume canonical/persisted data under governed access	Not assigned — TBD/Human Review
ARGUS, Memory, Agent & AI (WBS-19)	Security, IAM & Governance; Data, Database & Storage	Sandbox/security controls and Memory persistence both precede Agent/ARGUS activation	Not assigned — TBD/Human Review
Evidence, Provenance & Audit (WBS-20)	Data, Database & Storage; Security, IAM & Governance	Evidence persistence and custody controls precede claims requiring defensibility	Not assigned — TBD/Human Review

Functional Modules (WBS-21)	Data, Database & Storage; Evidence, Provenance & Audit; API, Integration & Connector	Domain rule evaluation requires canonical persistence, Evidence interface and integration inputs	Not assigned — TBD/Human Review
UX/UI & Design System (WBS-22)	API, Integration & Connector; ARGUS/Memory/Agent/AI; Functional Modules (for state presentation)	UX presentation follows the authoritative states/contracts it must render	Not assigned — TBD/Human Review
DevSecOps, Infrastructure & Deployment (WBS-23)	Foundation / Platform; Security, IAM & Governance	CI/CD and environment enablement can proceed in parallel with data/AI once platform/security foundations exist	Not assigned — TBD/Human Review
Testing, Verification & Acceptance (WBS-24)	All preceding workstreams (workstream-specific test entry per Ch.24)	Verification consumes the outputs of every other workstream it tests	Not assigned — TBD/Human Review

9.R RACI / Control Ownership

Table 15. Chapter 9 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Roadmap)	Consulted	Informed
Confirm sequencing/critical-path governance	Accountable Authority Class (to be assigned)	Sequencing Owner (to be assigned)	NCIE-001-016 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR17-9-1: Confirm sequencing/critical-path governance (Table 14; no calendar date assigned by this document).

Traceability: NCIE-001–016.

Parallelization & Workstream Coordination

Define safe parallel workstreams and coordination points.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Planning specification only; nothing in this chapter implies implementation work has started, tests have passed, blockers have cleared, or release/go-live is authorized.

Upstream: NCIE-001–016.

10.0 Salient Things to Note

PARALLELIZABLE ≠ INDEPENDENT; shared contracts/data/security gates remain explicit — parallel execution is safe only where shared dependencies are named, never assumed.

10.1 Parallel Workstreams and Shared Dependencies

PARALLELIZABLE ≠ INDEPENDENT — two workstreams with no direct predecessor relationship may still share a contract, a data dependency, or a security control; parallel execution is safe only where those shared dependencies are made explicit, per Chapter 8's dependency register.

10.2 Integration Points and Synchronization Gates

Integration points (where parallel workstreams' outputs must reconcile) and synchronization gates (explicit checkpoints before parallel work may proceed further) are named per workstream pair, so parallelization is a coordinated design choice, not an assumption.

10.R RACI / Control Ownership

Table 16. Chapter 10 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Roadmap)	Consulted	Informed
Confirm workstream structure and coordination authority	Accountable Authority Class (to be assigned)	Coordination Owner (to be assigned)	NCIE-001-016 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR17-10-1: Confirm workstream structure and coordination authority.

Traceability: NCIE-001–016.

Milestone & Stage-Gate Architecture

Define milestones and gates from architecture handover through go-live.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Planning specification only; nothing in this chapter implies implementation work has started, tests have passed, blockers have cleared, or release/go-live is authorized.

Upstream: NCIE-001–016.

11.0 Salient Things to Note

MILESTONE REACHED ≠ ACCEPTANCE GATE PASSED — no gate silently defaults to passed in the absence of an explicit approver decision.

11.1 Milestone and Gate Catalogue

MILESTONE REACHED ≠ ACCEPTANCE GATE PASSED — a milestone marks a planning checkpoint (e.g. 'foundation workstream work packages defined'); a stage gate additionally requires the evidence, approver authority and non-Blocking-status confirmation defined in Chapter 12 before the next stage may begin.

11.2 Blocked and Conditional Gate States

Every gate can be Open, Blocked (unmet evidence/HR item) or Conditionally Passed (approved with stated conditions); no gate silently defaults to passed in the absence of an explicit approver decision.

11.R RACI / Control Ownership

Table 17. Chapter 11 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Roadmap)	Consulted	Informed
Confirm milestone/gate catalogue	Accountable Authority Class (to be assigned)	Milestone/Gate Owner (to be assigned)	NCIE-001-016 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR17-11-1: Confirm milestone/gate catalogue.

Traceability: NCIE-001–016.

Entry / Exit Criteria by Stage

Define evidence-based entry/exit criteria for each stage.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Planning specification only; nothing in this chapter implies implementation work has started, tests have passed, blockers have cleared, or release/go-live is authorized.

Upstream: NCIE-001–016.

12.0 Salient Things to Note

Exit requires stated evidence; completion labels cannot substitute for evidence — Table 18 fixes the entry/exit evidence and approval authority for every stage from handover through go-live.

12.1 Entry Evidence and Exit Evidence

Every stage states what evidence is required to enter it and what evidence is required to exit it; exit requires stated evidence — completion labels ('done', 'complete') cannot substitute for the evidence itself.

12.2 Stage-Gate Matrix (directive item 66)

The consolidated matrix below maps each implementation stage to entry evidence, exit evidence, Blocking HR items, NCIE-016 tests, defects/residual gaps, approval authority and allowed next state.

Table 18. Stage-Gate Matrix.

Stage	Entry Evidence	Exit Evidence	Blocking HR Items	NCIE-016 Tests	Defects/Residual Gaps	Approval Authority	Allowed Next State
Architecture-to-implementation handover (Ch.13)	Chapter 3 baseline freeze confirmed	Handover package complete per Ch.13 contents list, credentials excluded	Per Chapter 4 intake	N/A at this stage	Tracked in Ch.34 RAID	NCIE-002 Ch.32 handover authority	Backlog population (Ch.14)
Work-package readiness (Ch.7, 37)	Handover package accepted	Definition of Ready met per Ch.37	Per work package	N/A at this stage	Tracked in Ch.34 RAID	Workstream Owner (Ch.15-23)	Work-package execution (downstream of this document)
Work-package completion (Ch.37)	Work started (downstream)	Definition of Done met per Ch.37	Per work package	Unit/integration per NCIE-016 Ch.5	Tracked in Ch.29 defect governance (NCIE-016)	Workstream Owner	Test-integration entry (Ch.24)
Test integration (Ch.24)	Definition of Done evidence	NCIE-016 RTM entries executed with recorded results	NCIE-016 HR16 items	Per NCIE-016 Ch.5 taxonomy	Per NCIE-016 Ch.29 defect register	NCIE-016 Chapter 31 independent reviewer	Acceptance-gate entry (Ch.38)
Acceptance gate (Ch.38)	Test results,	NCIE-008 acceptance	All applicable	All applicable	Zero unaddressed	NCIE-008 accounta	Release/cutover planning

	defect status, residual-risk record	ce decision recorded (Chapter 32 matrix of NCIE-016)	ble HR1-17 Blocking items	NCIE-016 tests	Blocking defects, or governed deferral	ble authority	(Ch.29)
Release/cutover /go-live (Ch.29)	Acceptanc e gate passed	Deploym ent receipt + health verificatio n (per NCIE-015 Ch.8)	NCIE-015 HR15 go-live items	N/A — operational, not a test stage	N/A	NCIE-015/NCIE -008 go-live authority	N/A — terminal for this document's scope

12.R RACI / Control Ownership

Table 19. Chapter 12 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Roadmap)	Consulted	Informed
Confirm entry/exit evidence and gate authority	Accountable Authority Class (to be assigned)	Entry/Exit Criteria Owner (to be assigned)	NCIE-001-016 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR17-12-1: Confirm entry/exit evidence and gate authority (Table 18).

Traceability: NCIE-001–016.

Architecture-to-Codex / Implementation Handover

Define controlled handover package to implementation/Codex teams.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Planning specification only; nothing in this chapter implies implementation work has started, tests have passed, blockers have cleared, or release/go-live is authorized.

Upstream: NCIE-002 Ch.32 architecture-to-Codex handover.

13.0 Salient Things to Note

HANDOVER PACKAGE ≠ CREDENTIAL TRANSFER; credentials/secrets are excluded from documentation handover — Table 20 maps every upstream document to its implementation package, test handover and operational ownership without ever referencing a credential value.

13.1 Handover Contents

The handover package references approved architecture, the WBS, RTM links, acceptance criteria and runbook/backlog/deployment/recovery references — it hands over pointers to controlled documents, not copies that could drift from their source.

13.2 Credential Exclusion

HANDOVER PACKAGE ≠ CREDENTIAL TRANSFER — no secret, password, token, private key or live credential is ever placed in a roadmap or handover artifact; where implementation needs credential access, the package references the NCIE-009/NCIE-015 credential-delivery mechanism, never the credential itself.

13.3 Architecture-to-Implementation Handover Matrix (directive item 67)

The consolidated matrix below maps each NCIE document to its implementation work packages, required artifacts, unresolved HR dependencies, test handover and downstream operational ownership.

Table 20. Architecture-to-Implementation Handover Matrix.

NCIE Document	Implementation Work Packages	Required Artifacts	Unresolved HR Dependencies	Test Handover (NCIE-016)	Downstream Operational Ownership (NCIE-015)
NCIE-003/014 (Data/Persistence)	WBS-17 work packages	Canonical mapping register, physical schema references	HR14 items per Chapter 4	NCIE-016 Ch.12 data/migration tests	NCIE-015 Ch.15 database operations
NCIE-009 (Security/IAM)	WBS-16 work packages	Zero-Trust control references, secrets interface	HR9 items per Chapter 4	NCIE-016 Ch.19 security tests	NCIE-015 Ch.13-14 IAM/secrets operations
NCIE-005/006/007 (ARGUS/Memory/Agent/AI)	WBS-19 work packages	Behavioral scenario corpus references, Gate A/B references	HR5/6/7 items per Chapter 4	NCIE-016 Ch.13-18 AI/Agent tests	NCIE-015 Ch.11, 17 Agent runtime/ARGUS operations
NCIE-010 (Evidence/Audit)	WBS-20 work packages	Evidence object/custody references	HR10 items per Chapter 4	NCIE-016 Ch.20 defensibility tests	NCIE-015 Ch.13 Evidence persistence operations
NCIE-013 (Functional	WBS-21 work	Domain rule/KPI framework	HR13 items	NCIE-016 Ch.21 domain	NCIE-015 Ch.18

Modules)	packages	references	per Chapter 4	scenario tests	functional- module operations
NCIE-012 (UX/UI)	WBS-22 work packages	Design- system/compone nt references	HR12 items per Chapter 4	NCIE-016 Ch.22 UX/accessibilit y tests	NCIE-015 Ch.19 frontend delivery operations
NCIE-015 (DevSecOps)	WBS-23 work packages	CI/CD/envIRONme nt references	HR15 items per Chapter 4	NCIE-016 Ch.23 pipeline tests	NCIE-015 (self- operating)

13.R RACI / Control Ownership

Table 21. Chapter 13 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Roadmap)	Consulted	Informed
Confirm implementation/Codex handover authority and package contents	Accountable Authority Class (to be assigned)	Handover Owner (to be assigned)	NCIE-001-016 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR17-13-1: Confirm implementation/Codex handover authority and package contents (Table 20).

Traceability: NCIE-002 Ch.32 architecture-to-Codex handover.

Implementation Backlog & Prioritization

Define backlog structure and prioritization governance.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Planning specification only; nothing in this chapter implies implementation work has started, tests have passed, blockers have cleared, or release/go-live is authorized.

Upstream: NCIE-001–016.

14.0 Salient Things to Note

Backlog priority cannot override Blocking Human Review, security or governance gates — a high-priority item with an open Blocking HR item remains blocked regardless of its priority rank.

14.1 Backlog Item Fields

Every backlog item carries its priority rationale, dependencies (per Chapter 8), risks (per Chapter 34), HR blockers (per Chapter 4) and readiness state (per Chapter 37), so prioritization decisions are made with full visibility of what actually blocks an item.

14.2 Priority Cannot Override Governance/Security Blockers

Backlog priority cannot override Blocking Human Review, security or governance gates — a high-priority item with an open Blocking HR item remains blocked; priority affects sequencing among unblocked items only.

14.R RACI / Control Ownership

Table 22. Chapter 14 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Roadmap)	Consulted	Informed
Confirm prioritization authority/criteria	Accountable Authority Class (to be assigned)	Backlog Owner (to be assigned)	NCIE-001-016 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR17-14-1: Confirm prioritization authority/criteria.

Traceability: NCIE-001–016.

Foundation / Platform Workstream

Sequence shared foundation/platform capabilities.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Planning specification only; nothing in this chapter implies implementation work has started, tests have passed, blockers have cleared, or release/go-live is authorized.

Upstream: NCIE-001–016.

15.0 Salient Things to Note

Foundation work must not pre-empt unresolved technology/infrastructure decisions — sequencing proceeds on the workstream order (Table 14) without committing to any product NCIE-004 has not already fixed.

15.1 Core Services, Shared Runtime and Identity Foundations

Foundation work must not pre-empt unresolved technology/infrastructure decisions — core services, shared runtime and identity foundations are sequenced as the first logical workstream (Table 14) without committing to a specific product NCIE-004 has not already fixed.

15.2 Observability Foundations

Observability foundations are sequenced early since NCIE-015 Chapter 20 and NCIE-016 Chapter 27 both depend on a working telemetry substrate existing before their own verification can proceed meaningfully.

15.R RACI / Control Ownership

Table 23. Chapter 15 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Roadmap)	Consulted	Informed
Confirm foundation technology decisions that block sequencing	Accountable Authority Class (to be assigned)	Foundation Workstream Owner (to be assigned)	NCIE-004 owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED] (Blocking) HR17-15-1: Confirm foundation technology decisions that block sequencing.

Traceability: NCIE-001–016.

Security, IAM & Governance Workstream

Sequence NCIE-008/009 governance, IAM, Zero Trust, secrets, network, privileged access and audit prerequisites as enabling dependencies.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Planning specification only; nothing in this chapter implies implementation work has started, tests have passed, blockers have cleared, or release/go-live is authorized.

Upstream: NCIE-001–016.

16.0 Salient Things to Note

Security/governance controls are enabling dependencies, not post-build decoration — this workstream is sequenced immediately after Foundation/Platform because every later workstream depends on it.

16.1 Governance, IAM and Zero-Trust Sequencing

Security/governance controls are enabling dependencies, not post-build decoration — Table 14 sequences this workstream immediately after Foundation/Platform specifically because every later workstream depends on the access control it establishes.

16.2 Secrets, Network, Privileged Access and Audit Prerequisites

Secrets delivery, network segmentation, privileged-access controls and audit-logging prerequisites (NCIE-009 Chapters 12-19, 28) are sequenced as a block, since a later workstream consuming any one of them without the others creates an incomplete security posture.

16.R RACI / Control Ownership

Table 24. Chapter 16 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Roadmap)	Consulted	Informed
Confirm governance/security implementation prerequisites	Accountable Authority Class (to be assigned)	Security/Governance Workstream Owner (to be assigned)	NCIE-008/009 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED] (Blocking) HR17-16-1: Confirm governance/security implementation prerequisites.

Traceability: NCIE-001–016.

Data, Database & Storage Workstream

Sequence NCIE-003/014 canonical data, physical persistence, indexes, migration, backup/DR and reconciliation while preserving unresolved HR14 decisions.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Planning specification only; nothing in this chapter implies implementation work has started, tests have passed, blockers have cleared, or release/go-live is authorized.

Upstream: NCIE-001–016.

17.0 Salient Things to Note

Physical persistence must remain subordinate to canonical semantics and unresolved HR14 decisions — this workstream sequences NCIE-014's realization without resolving any of its own open items.

17.1 Canonical Schemas, Stores and Indexes

Physical persistence must remain subordinate to canonical semantics and unresolved HR14 decisions — this workstream sequences NCIE-014's physical realization exactly as NCIE-014 defines it, without resolving any of NCIE-014's own open technology-selection items.

17.2 Migration and Backup/DR Foundations

Migration and backup/DR foundations are sequenced before Functional Modules (WBS-21) and Evidence/Audit (WBS-20), since both later workstreams assume a working, recoverable persistence layer beneath them.

17.R RACI / Control Ownership

Table 25. Chapter 17 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Roadmap)	Consulted	Informed
Confirm persistence technology/HR14 dependencies	Accountable Authority Class (to be assigned)	Data/Storage Workstream Owner (to be assigned)	NCIE-003/014 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED] (Blocking) HR17-17-1: Confirm persistence technology/HR14 dependencies.

Traceability: NCIE-001–016.

API, Integration & Connector Workstream

Sequence NCIE-011 API gateway, events, schemas, connectors and external integration with source discovery, security and contract dependencies.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Planning specification only; nothing in this chapter implies implementation work has started, tests have passed, blockers have cleared, or release/go-live is authorized.

Upstream: NCIE-001–016.

18.0 Salient Things to Note

Connector/provider build sequencing depends on source discovery and current authorization — a connector cannot be built to a contract that has not yet been discovered or certified.

18.1 Gateway, Events and Schemas

Connector/provider build sequencing depends on source discovery and current authorization — gateway/event/schema work packages are sequenced only after the Security/Governance workstream (Chapter 16) and Data/Storage workstream (Chapter 17) since integration both authorizes and moves canonical data.

18.2 External Connectors and Provider Contracts

External connector and provider-contract work packages are cross-referenced to Chapter 27's external-systems sequencing, since a connector cannot be built to a contract that has not yet been discovered or certified.

18.R RACI / Control Ownership

Table 26. Chapter 18 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Roadmap)	Consulted	Informed
Confirm external/provider discovery and connector prerequisites	Accountable Authority Class (to be assigned)	Integration Workstream Owner (to be assigned)	NCIE-011 owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [SOURCE/WORKFLOW DISCOVERY REQUIRED] (Blocking) HR17-18-1: Confirm external/provider discovery and connector prerequisites.

Traceability: NCIE-001–016.

ARGUS, Memory, Agent & AI Workstream

Sequence ARGUS, Memory, Agent, Model, Tool and VPF capabilities.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Planning specification only; nothing in this chapter implies implementation work has started, tests have passed, blockers have cleared, or release/go-live is authorized.

Upstream: NCIE-001–016.

19.0 Salient Things to Note

AI/Agent capability sequencing cannot bypass Gate A/B, VPF, sandbox or Human-Primary controls — a Tool Gateway work package is never marked ready until its Gate A/VPF prerequisite is itself satisfied.

19.1 ARGUS, Memory/Context and Agent Factory/Runtime

AI/Agent capability sequencing cannot bypass Gate A/B, VPF, sandbox or Human-Primary controls — this workstream sequences ARGUS, Memory/Context and Agent Factory/runtime only after the Security/Governance (Chapter 16) and Data/Storage (Chapter 17) workstreams, since sandbox isolation and Memory persistence both precede activation.

19.2 Model Gateway, Tool Gateway and VPF

Model Gateway, Tool Gateway and VPF work packages are sequenced with explicit Gate A/B checkpoints inline — a work package building Tool Gateway capability is not marked ready until its corresponding VPF/Gate A prerequisite from NCIE-007 is itself satisfied.

19.R RACI / Control Ownership

Table 27. Chapter 19 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Roadmap)	Consulted	Informed
Confirm AI/Agent implementation prerequisites	Accountable Authority Class (to be assigned)	AI/Agent Workstream Owner (to be assigned)	NCIE-005/006/007 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED] (Blocking) HR17-19-1: Confirm AI/Agent implementation prerequisites.

Traceability: NCIE-001–016.

Evidence, Provenance & Audit Workstream

Sequence NCIE-010 Evidence, custody, provenance, audit and Decision-Time capabilities before dependent defensibility claims.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Planning specification only; nothing in this chapter implies implementation work has started, tests have passed, blockers have cleared, or release/go-live is authorized.

Upstream: NCIE-001–016.

20.0 Salient Things to Note

Evidence/audit implementation must precede claims that require traceability/defensibility — this workstream is sequenced before Functional Modules precisely because Evidence-linked domain outputs require it first.

20.1 Evidence Objects, Custody and Provenance

Evidence/audit implementation must precede claims that require traceability/defensibility — this workstream is sequenced before Functional Modules (Chapter 21) specifically because domain-module Evidence-linked outputs (per Table 20) require a working Evidence interface first.

20.2 Audit and Decision-Time Sequencing

Audit-trail and Decision-Time capability work packages are sequenced together, since Decision-Time correctness (what a Decision actually saw) depends on the audit trail existing to reconstruct it.

20.R RACI / Control Ownership

Table 28. Chapter 20 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Roadmap)	Consulted	Informed
Confirm Evidence/audit minimum enabling sequence	Accountable Authority Class (to be assigned)	Evidence/Audit Workstream Owner (to be assigned)	NCIE-010 owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR17-20-1: Confirm Evidence/audit minimum enabling sequence.

Traceability: NCIE-001–016.

Functional Modules Workstream

Sequence approved Functional Modules and domain dependencies.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Planning specification only; nothing in this chapter implies implementation work has started, tests have passed, blockers have cleared, or release/go-live is authorized.

Upstream: NCIE-001–016.

21.0 Salient Things to Note

Domain module implementation cannot invent unresolved rules/KPIs/thresholds — every rule/KPI/threshold decision is deferred to the same HR13 dependency Chapter 4 already carries.

21.1 Domain Module Sequencing

Domain module implementation cannot invent unresolved rules/KPIs/thresholds — this workstream sequences the Network, Traffic, Revenue, Mobile Money, SIM, Anti-Fraud and REWS modules per NCIE-013's own Chapter structure, deferring every rule/KPI/threshold to the Chapter 4 HR13 dependency it already carries.

21.2 Cross-Domain and Evidence Dependencies

Cross-domain correlation work packages are sequenced after individual domain modules are ready, and every Evidence-linked module output depends on the Chapter 20 Evidence workstream having delivered its interface first (per Table 12's DEP-20-21 entry).

21.R RACI / Control Ownership

Table 29. Chapter 21 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Roadmap)	Consulted	Informed
Confirm domain-rule/KPI dependencies	Accountable Authority Class (to be assigned)	Domain Modules Workstream Owner (to be assigned)	NCIE-013 owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR17-21-1: Confirm domain-rule/KPI dependencies.

Traceability: NCIE-001–016.

UX/UI & Design System Workstream

Sequence UX/design-system implementation against upstream states/contracts.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Planning specification only; nothing in this chapter implies implementation work has started, tests have passed, blockers have cleared, or release/go-live is authorized.

Upstream: NCIE-001–016.

22.0 Salient Things to Note

UX work may proceed with prototypes where allowed, but presentation cannot outrun authoritative semantics — any work package presenting live governance/security/Evidence state is sequenced after the workstream that actually produces it.

22.1 Design System, Shell and ARGUS UX

UX work may proceed with prototypes where allowed, but presentation cannot outrun authoritative semantics — design-system and shell work packages may start early using mock data/states, but any work package presenting live governance/security/Evidence state is sequenced after the workstream that actually produces that state.

22.2 Governance/Security/Evidence States and Accessibility

Work packages that present governance, security or Evidence state are cross-referenced to Chapters 16 and 20 as hard predecessors (not merely 'preferred'); accessibility work packages proceed per NCIE-012's own standard, deferred to Human Review for the specific conformance level.

22.R RACI / Control Ownership

Table 30. Chapter 22 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Roadmap)	Consulted	Informed
Confirm UX implementation/prototype dependencies	Accountable Authority Class (to be assigned)	UX Workstream Owner (to be assigned)	NCIE-012 owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Non-Blocking) HR17-22-1: Confirm UX implementation/prototype dependencies; does not block the state-presentation predecessor discipline already defined in the interim.

Traceability: NCIE-001–016.

DevSecOps, Infrastructure & Deployment Workstream

Sequence DevSecOps/infrastructure/deployment enablement.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Planning specification only; nothing in this chapter implies implementation work has started, tests have passed, blockers have cleared, or release/go-live is authorized.

Upstream: NCIE-001–016.

23.0 Salient Things to Note

Deployment automation cannot self-authorize release or production acceptance — every pipeline gate this workstream builds remains subject to the Chapter 38 acceptance-gate integration.

23.1 CI/CD, Environments and IaC

Deployment automation cannot self-authorize release or production acceptance — CI/CD, environment and IaC work packages proceed largely in parallel with Data/AI workstreams once Foundation/Platform and Security/Governance exist (per Table 14), but every pipeline gate they build remains subject to the Chapter 38 acceptance-gate integration.

23.2 Runtime, Observability, Incident/Runbooks and Deployment

Runtime, observability, incident/runbook and deployment work packages are cross-referenced to NCIE-015's own chapter structure so this roadmap never restates operational architecture NCIE-015 already owns — it only sequences building it.

23.R RACI / Control Ownership

Table 31. Chapter 23 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Roadmap)	Consulted	Informed
Confirm infrastructure/platform prerequisites	Accountable Authority Class (to be assigned)	DevSecOps Workstream Owner (to be assigned)	NCIE-015 owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED] (Blocking) HR17-23-1: Confirm infrastructure/platform prerequisites.

Traceability: NCIE-001–016.

Testing, Verification & Acceptance Workstream

Integrate NCIE-016 verification into build sequencing.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Planning specification only; nothing in this chapter implies implementation work has started, tests have passed, blockers have cleared, or release/go-live is authorized.

Upstream: NCIE-016 approved verification architecture.

24.0 Salient Things to Note

BUILD COMPLETE ≠ TEST PASSED; TEST PASSED ≠ PRODUCTION ACCEPTANCE — this workstream integrates NCIE-016's architecture into the sequence exactly as defined, adding no new test requirement.

24.1 RTM, Test Cases, Environments and Evidence

This workstream integrates NCIE-016's RTM, test-case, environment and evidence architecture into the build sequence exactly as NCIE-016 defines it, adding no new test requirement — BUILD COMPLETE ≠ TEST PASSED remains true at every point in this sequencing.

24.2 Defects, Acceptance and Regression

TEST PASSED ≠ PRODUCTION ACCEPTANCE — defect handling follows NCIE-016 Chapter 29 exactly; acceptance follows NCIE-016 Chapter 32's matrix, consumed (not reproduced) by this document's Chapter 38; regression testing is sequenced after every change per NCIE-016 Chapter 28.

24.R RACI / Control Ownership

Table 32. Chapter 24 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Roadmap)	Consulted	Informed
Confirm NCIE-016 execution/acceptance sequencing	Accountable Authority Class (to be assigned)	Testing Integration Owner (to be assigned)	NCIE-016 owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR17-24-1: Confirm NCIE-016 execution/acceptance sequencing.

Traceability: NCIE-016 approved verification architecture.

Environment & Configuration Sequencing

Define dev/test/stage/prod environment and configuration sequencing, data classes, secrets boundaries and promotion dependencies without inventing topology.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Planning specification only; nothing in this chapter implies implementation work has started, tests have passed, blockers have cleared, or release/go-live is authorized.

Upstream: NCIE-001–016.

25.0 Salient Things to Note

Environment promotion and configuration deployment remain governed and versioned — no topology is invented; the actual topology remains an NCIE-015 Human Review decision this roadmap only sequences around.

25.1 Environment Sequencing

Environment promotion and configuration deployment remain governed and versioned — dev, test, stage and prod environments are sequenced per NCIE-015 Chapter 3's own environment model, with each higher environment's readiness gated on the lower environment's exit criteria (Chapter 12).

25.2 Data Classes, Secrets Boundaries and Promotion Dependencies

Data classification and secrets boundaries follow NCIE-009 policy exactly at every environment tier; no environment topology is invented by this document — the actual topology remains an NCIE-015 Human Review decision this roadmap only sequences around.

25.R RACI / Control Ownership

Table 33. Chapter 25 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Roadmap)	Consulted	Informed
Confirm environment topology/configuration authority	Accountable Authority Class (to be assigned)	Environment Sequencing Owner (to be assigned)	NCIE-009/015 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED] (Blocking) HR17-25-1: Confirm environment topology/configuration authority (no topology invented by this document).

Traceability: NCIE-001–016.

Data Migration & Initial-Load Sequencing

Sequence legacy/initial data migration and reconciliation.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Planning specification only; nothing in this chapter implies implementation work has started, tests have passed, blockers have cleared, or release/go-live is authorized.

Upstream: NCIE-001–016.

26.0 Salient Things to Note

MIGRATION SUCCESS ≠ RECONCILIATION COMPLETE; initial load needs semantic acceptance — cutover is sequenced only after reconciliation authority signs off, never after a copy operation's success report alone.

26.1 Source Inventory, Mapping and Transform

This workstream sequences NCIE-014 Chapter 26's migration architecture (source inventory, mapping, transform, backfill) without re-defining any of its reconciliation requirements; MIGRATION SUCCESS ≠ RECONCILIATION COMPLETE remains true throughout the sequence.

26.2 Reconcile and Cutover

Initial load needs semantic acceptance, not just technical completion — cutover to the new system of record is sequenced only after reconciliation authority (per NCIE-014 Chapter 26) signs off, never merely after the copy operation reports success.

26.R RACI / Control Ownership

Table 34. Chapter 26 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Roadmap)	Consulted	Informed
Confirm migration sources/cutover authority	Accountable Authority Class (to be assigned)	Migration Sequencing Owner (to be assigned)	NCIE-014 owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR17-26-1: Confirm migration sources/cutover authority.

Traceability: NCIE-001–016.

External Systems / Provider Integration Sequencing

Sequence provider/external-system discovery, onboarding and adapters.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Planning specification only; nothing in this chapter implies implementation work has started, tests have passed, blockers have cleared, or release/go-live is authorized.

Upstream: NCIE-001–016.

27.0 Salient Things to Note

External availability ≠ integration readiness; contracts, security and discovery dependencies remain explicit — technical reachability alone never substitutes for contract/security certification.

27.1 Source Discovery and Contracts

External availability ≠ integration readiness — an external system being technically reachable does not mean its contract, security posture and data semantics have been discovered and certified; discovery and contract work packages precede adapter-building work packages.

27.2 Credentials Interface, Certification and Fallback

Credentials interface (never credential values — per Chapter 13's exclusion) and certification/fallback work packages are cross-referenced to Chapter 33's procurement tracking, since an external dependency's readiness is itself evidence-based, not presumed.

27.R RACI / Control Ownership

Table 35. Chapter 27 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Roadmap)	Consulted	Informed
Confirm external systems/provider sequencing	Accountable Authority Class (to be assigned)	External Systems Owner (to be assigned)	NCIE-001-016 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [SOURCE/WORKFLOW DISCOVERY REQUIRED] (Blocking) HR17-27-1: Confirm external systems/provider sequencing.

Traceability: NCIE-001–016.

Operational Readiness & Runbook Sequencing

Sequence runbooks, ownership, monitoring and operational readiness.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Planning specification only; nothing in this chapter implies implementation work has started, tests have passed, blockers have cleared, or release/go-live is authorized.

Upstream: NCIE-001–016.

28.0 Salient Things to Note

Runbooks/owners/monitoring are readiness dependencies, not documentation afterthoughts — go-live needs all of them visible together at the acceptance gate, never a partial subset.

28.1 Service Ownership and Runbooks

Runbooks/owners/monitoring are readiness dependencies, not documentation afterthoughts — this workstream sequences NCIE-015 Chapter 24's runbook architecture and Chapter 33's readiness checklist to complete before the Chapter 29 go-live dependency plan can proceed.

28.2 Monitoring, Alerting, Backup/DR and Support Readiness

Monitoring, alerting, backup/DR and support-readiness work packages are sequenced together since a go-live decision needs all of them, not a partial subset, visible at the Chapter 12 acceptance gate.

28.R RACI / Control Ownership

Table 36. Chapter 28 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Roadmap)	Consulted	Informed
Confirm operational ownership/runbook readiness authority	Accountable Authority Class (to be assigned)	Operational Readiness Owner (to be assigned)	NCIE-015 owner	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR17-28-1: Confirm operational ownership/runbook readiness authority.

Traceability: NCIE-001–016.

Release, Cutover & Go-Live Dependency Plan

Define release/cutover/go-live dependency plan without self-authorizing go-live.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Planning specification only; nothing in this chapter implies implementation work has started, tests have passed, blockers have cleared, or release/go-live is authorized.

Upstream: NCIE-001–016.

29.0 Salient Things to Note

DEPLOYED ≠ APPROVED; go-live requires authorized acceptance gates — this plan sequences release/cutover dependencies without adding any new authorization path of its own.

29.1 Release Candidate, Gates and Migration

DEPLOYED ≠ APPROVED — this workstream sequences release-candidate, required-gate, migration and deployment dependencies exactly per NCIE-015 Chapter 8's Release-Control Matrix, adding no new authorization path of its own.

29.2 Smoke Verification, Acceptance and Rollback

Go-live requires authorized acceptance gates — smoke verification (per NCIE-016 Chapter 38) and the Chapter 38 acceptance-gate integration both precede any go-live dependency being marked satisfied; rollback dependencies are sequenced identically to the forward path so a reversal is never improvised.

29.R RACI / Control Ownership

Table 37. Chapter 29 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Roadmap)	Consulted	Informed
Confirm release/cutover/go-live authority	Accountable Authority Class (to be assigned)	Release/Go-Live Owner (to be assigned)	NCIE-008/015 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR17-29-1: Confirm release/cutover/go-live authority.

Traceability: NCIE-001–016.

Rollback, Recovery & Contingency Planning

Define rollback/recovery contingencies as roadmap dependencies.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Planning specification only; nothing in this chapter implies implementation work has started, tests have passed, blockers have cleared, or release/go-live is authorized.

Upstream: NCIE-001–016.

30.0 Salient Things to Note

Rollback/recovery planning does not prove recovery capability; NCIE-016 evidence remains required — planning work and validated capability remain distinct throughout this chapter.

30.1 Rollback Triggers and Recovery Paths

Rollback/recovery planning does not prove recovery capability — this chapter sequences the planning work (defining triggers, paths, communications) that precedes NCIE-016 Chapter 24's actual recovery-validation testing; planning and validated capability remain distinct.

30.2 Data/State Reconciliation, Communications and Evidence

Contingency plans require data/state reconciliation and communications work packages alongside the technical rollback mechanism, since a technically successful rollback that leaves stale state or uninformed stakeholders is not a complete contingency response.

30.R RACI / Control Ownership

Table 38. Chapter 30 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Roadmap)	Consulted	Informed
Confirm rollback/recovery decision authority	Accountable Authority Class (to be assigned)	Contingency Planning Owner (to be assigned)	NCIE-014/015 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR17-30-1: Confirm rollback/recovery decision authority.

Traceability: NCIE-001–016.

Resource, Skill & Ownership Planning

Define required skills/roles/ownership without inventing named staff.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Planning specification only; nothing in this chapter implies implementation work has started, tests have passed, blockers have cleared, or release/go-live is authorized.

Upstream: NCIE-001–016.

31.0 Salient Things to Note

Resource plan describes required capabilities; it must not fabricate people, budgets or availability — Table 39's Capacity Assumption column reads Not sourced throughout because no approved source supplies one.

31.1 Role/Skill Categories and Responsibility

Resource plan describes required capabilities; it must not fabricate people, budgets or availability — every workstream's required role/skill categories are stated generically (e.g. 'Security Engineer', 'Data Migration Specialist'), never as a named individual or a headcount commitment.

31.2 Resource/Skill Matrix (directive item 69)

The consolidated matrix below maps workstreams to required role/skill categories and capacity assumptions without named individuals, salaries, headcounts or availability figures unless explicitly sourced from an approved document (none is currently supplied).

Table 39. Resource/Skill Matrix.

Workstream	Required Role/Skill Categories	Capacity Assumption
Foundation / Platform (WBS-15)	Platform/Infrastructure Engineer, Site Reliability Engineer	Not sourced — TBD/Human Review
Security, IAM & Governance (WBS-16)	Security Engineer, IAM Engineer, Governance Analyst	Not sourced — TBD/Human Review
Data, Database & Storage (WBS-17)	Database Engineer, Data Migration Specialist	Not sourced — TBD/Human Review
API, Integration & Connector (WBS-18)	Integration Engineer, API Developer	Not sourced — TBD/Human Review
ARGUS, Memory, Agent & AI (WBS-19)	AI/ML Engineer, Agent Systems Engineer, Conversational Design Specialist	Not sourced — TBD/Human Review
Evidence, Provenance & Audit (WBS-20)	Evidence/Compliance Engineer, Audit Systems Engineer	Not sourced — TBD/Human Review
Functional Modules (WBS-21)	Domain/Business Logic Engineer, Data Analyst	Not sourced — TBD/Human Review
UX/UI & Design System (WBS-22)	UX/UI Designer, Frontend Engineer, Accessibility Specialist	Not sourced — TBD/Human Review
DevSecOps, Infrastructure & Deployment (WBS-23)	DevSecOps Engineer, Release Engineer	Not sourced — TBD/Human Review
Testing, Verification & Acceptance (WBS-24)	QA/Test Engineer, Security Test Specialist	Not sourced — TBD/Human Review

31.R RACI / Control Ownership

Table 40. Chapter 31 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Roadmap)	Consulted	Informed
Confirm skill/resource availability assumptions	Accountable Authority Class (to be assigned)	Resource Planning Owner (to be assigned)	NCIE-001-016 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR17-31-1: Confirm skill/resource availability assumptions (Table 39; no headcount/budget invented by this document).

Traceability: NCIE-001–016.

Estimate, Duration & Schedule Governance

Govern estimates/durations/schedules without fabricating commitments.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Planning specification only; nothing in this chapter implies implementation work has started, tests have passed, blockers have cleared, or release/go-live is authorized.

Upstream: NCIE-001–016.

32.0 Salient Things to Note

Dates/durations require Human/source authority; uncertainty and assumptions remain explicit — no calendar date, duration, budget or delivery commitment appears anywhere in this specification.

32.1 Estimate Basis, Range and Confidence

Dates/durations require Human/source authority; uncertainty and assumptions remain explicit — any estimate this document might reference states its basis, range and confidence level rather than presenting a single fabricated number as fact.

32.2 Calendar Authority and the TBD Default

If no authoritative dates/durations exist in an approved source, they are left explicitly TBD / Human Review rather than estimated by this document — consistent with directive item 9, no calendar date, duration, budget, or delivery commitment appears anywhere in this specification.

32.R RACI / Control Ownership

Table 41. Chapter 32 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Roadmap)	Consulted	Informed
Confirm estimate/schedule authority and planning horizon	Accountable Authority Class (to be assigned)	Schedule Governance Owner (to be assigned)	NCIE-001-016 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR17-32-1: Confirm estimate/schedule authority and planning horizon (no date/duration invented by this document).

Traceability: NCIE-001–016.

Procurement / External Dependency Tracking

Track procurement/vendor/provider/external prerequisites.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Planning specification only; nothing in this chapter implies implementation work has started, tests have passed, blockers have cleared, or release/go-live is authorized.

Upstream: NCIE-001–016.

33.0 Salient Things to Note

Procurement/provider dependency status must be evidence-based, not presumed — no lead time is invented; unsourced items read 'lead time not sourced' rather than an assumed duration.

33.1 Procurement Item, Provider and Contract

Procurement/provider dependency status must be evidence-based, not presumed — each tracked item states the provider, the contract/certification status and the specific work package it gates (cross-referenced to Chapter 27's external-systems sequencing).

33.2 Lead-Time Evidence, Blocker and Fallback

No procurement lead time is invented by this document; where lead-time evidence is not supplied, the item's blocker state reads 'lead time not sourced' rather than an assumed duration, and a fallback path is stated wherever one exists.

33.R RACI / Control Ownership

Table 42. Chapter 33 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Roadmap)	Consulted	Informed
Confirm procurement/provider dependencies	Accountable Authority Class (to be assigned)	Procurement Tracking Owner (to be assigned)	NCIE-001-016 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [SOURCE/WORKFLOW DISCOVERY REQUIRED] (Blocking) HR17-33-1: Confirm procurement/provider dependencies (no lead time/contract invented by this document).

Traceability: NCIE-001–016.

Risk, Assumption, Issue & Dependency (RAID) Register

Define RAID governance across implementation.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Planning specification only; nothing in this chapter implies implementation work has started, tests have passed, blockers have cleared, or release/go-live is authorized.

Upstream: NCIE-001–016.

34.0 Salient Things to Note

RAID status is not a substitute for resolution; accepted risk requires authorized governance — Table 43 shows every illustrative entry as Identified, never silently upgraded to accepted or closed without an owner-authority act.

34.1 Risk, Assumption, Issue and Dependency Categories

RAID status is not a substitute for resolution; accepted risk requires authorized governance — every RAID entry carries an owner-authority reference, and an 'accepted' risk is never indistinguishable from a resolved or closed one in the record.

34.2 RAID Register (directive item 70)

The consolidated register below defines the implementation RAID with evidence-based status and explicit separation among identified, mitigated, accepted, resolved and closed states, illustratively — the full register populates as each workstream's planning proceeds.

Table 43. Implementation RAID Register (illustrative structure).

RAID ID	Type	Description	Owner Authority	Mitigation/Response	Status
RAID-1	Risk	Unresolved HR9 security items may block Security/Governance workstream start (WBS-16)	NCIE-009 accountable authority	Escalate per Chapter 4 intake	Identified
RAID-2	Assumption	Foundation/Platform technology selection will be confirmed before Chapter 15 work packages are scheduled	Foundation Workstream Owner	Track via HR17-15-1	Identified
RAID-3	Issue	No procurement lead-time evidence currently supplied for external provider onboarding (Ch.27, 33)	Procurement Tracking Owner	Await source evidence	Identified
RAID-4	Dependency	Functional Modules workstream (WBS-21) depends on both Data/Storage (WBS-17) and Evidence/Audit (WBS-20) completion	Domain Modules Workstream Owner	Tracked in Table 12 dependency register	Identified
RAID-5	Risk	Resource/skill	Accountable	Escalate per HR17-	Identified

availability is entirely Authority Class 31-1
unsourced (Table
39), which may block
any workstream's
actual scheduling

34.R RACI / Control Ownership

Table 44. Chapter 34 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Roadmap)	Consulted	Informed
Confirm RAID ownership/risk-acceptance authority	Accountable Authority Class (to be assigned)	RAID Register Owner (to be assigned)	NCIE-001-016 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR17-34-1: Confirm RAID ownership/risk-acceptance authority (Table 43).

Traceability: NCIE-001–016.

Change Control & Roadmap Re-Baselining

Define controlled re-baselining when dependencies/architecture change.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Planning specification only; nothing in this chapter implies implementation work has started, tests have passed, blockers have cleared, or release/go-live is authorized.

Upstream: NCIE-001–016.

35.0 Salient Things to Note

CHANGE IN SEQUENCE ≠ CHANGE IN ARCHITECTURE; architecture change returns to upstream governance — re-baselining is never a silent overwrite of the prior baseline.

35.1 Change Trigger and Impact Assessment

CHANGE IN SEQUENCE ≠ CHANGE IN ARCHITECTURE — reordering work packages in response to a new dependency discovery is a sequencing change; it never itself implies an upstream architecture change, which returns to the relevant NCIE-001–016 document's own governance instead.

35.2 Approval, New Baseline and History

A confirmed change updates Chapter 3's freeze register to a new baseline, with the prior baseline preserved historically and the change traceable to its trigger and approval — re-baselining is never a silent overwrite.

35.R RACI / Control Ownership

Table 45. Chapter 35 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Roadmap)	Consulted	Informed
Confirm re-baselining/change authority	Accountable Authority Class (to be assigned)	Change Control Owner (to be assigned)	NCIE-001-016 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR17-35-1: Confirm re-baselining/change authority.

Traceability: NCIE-001–016.

Implementation Evidence & Progress Reporting

Define implementation-progress evidence and reporting.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Planning specification only; nothing in this chapter implies implementation work has started, tests have passed, blockers have cleared, or release/go-live is authorized.

Upstream: NCIE-001–016.

36.0 Salient Things to Note

Progress percentages/status must be evidence-backed and distinguish implementation from acceptance — a reader can never mistake 'work complete' for 'tested and accepted' in this reporting model.

36.1 Status Concepts and Artifacts

Progress percentages/status must be evidence-backed and distinguish implementation from acceptance — status is reported using the Chapter 71-item concept states (Not Started, Ready, Blocked, In Progress, Implemented/Work Complete, Test Pending, Tested, Acceptance Pending, Accepted) defined in the Implementation-Status Model, never a bare percentage.

36.2 Test State, Acceptance State and Anti-Fabrication

A status report never claims a percentage-complete figure without the underlying evidence artifact it derives from; test state and acceptance state are reported separately from implementation state so a reader cannot mistake 'work complete' for 'tested and accepted'.

36.R RACI / Control Ownership

Table 46. Chapter 36 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Roadmap)	Consulted	Informed
Confirm progress-reporting evidence/status rules	Accountable Authority Class (to be assigned)	Progress Reporting Owner (to be assigned)	NCIE-001-016 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR17-36-1: Confirm progress-reporting evidence/status rules.

Traceability: NCIE-001–016.

Definition of Ready / Definition of Done

Define Definition of Ready/Done for work packages without equating Done to acceptance.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Planning specification only; nothing in this chapter implies implementation work has started, tests have passed, blockers have cleared, or release/go-live is authorized.

Upstream: NCIE-001–016.

37.0 Salient Things to Note

Definition of Done is work-package completion, not automatic institutional acceptance — Done means outputs exist and pass their own evidence requirement, nothing more.

37.1 Definition of Ready

A work package is Ready when its dependencies (Chapter 8), inputs and owner role (Chapter 7) are all confirmed available — Ready is a scheduling precondition, not a quality judgment about the work itself.

37.2 Definition of Done and the Acceptance Boundary

Definition of Done is work-package completion, not automatic institutional acceptance — Done means the stated outputs exist and pass the work package's own Chapter 7 evidence requirement; explicit exclusions and any residual blockers are stated rather than implied absent.

37.R RACI / Control Ownership

Table 47. Chapter 37 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Roadmap)	Consulted	Informed
Confirm Definition of Ready/Done	Accountable Authority Class (to be assigned)	DoR/DoD Owner (to be assigned)	NCIE-001-016 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR17-37-1: Confirm Definition of Ready/Done.

Traceability: NCIE-001–016.

Acceptance-Gate & Sign-Off Integration

Integrate Human/Governance/test acceptance gates into roadmap states.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Planning specification only; nothing in this chapter implies implementation work has started, tests have passed, blockers have cleared, or release/go-live is authorized.

Upstream: NCIE-001–016.

38.0 Salient Things to Note

IMPLEMENTATION EVIDENCE ≠ ACCEPTANCE EVIDENCE UNLESS GOVERNED AS SUCH — evidence becomes acceptance evidence only when the accountable authority actually treats it as such at the gate.

38.1 Gate, Required Tests and Defects

Acceptance gates bind target/scope/version and consume NCIE-016 evidence plus Human/Governance authority — this chapter integrates, rather than duplicates, NCIE-016 Chapter 32's Acceptance/Sign-Off Matrix, treating it as the authoritative source for what evidence an acceptance decision requires.

38.2 HR Blockers, Approver and Conditions

IMPLEMENTATION EVIDENCE ≠ ACCEPTANCE EVIDENCE UNLESS GOVERNED AS SUCH — implementation evidence (Chapter 36) becomes acceptance evidence only when the accountable authority actually treats it as such at the gate; the approver's identity and any stated conditions are recorded with the same precision as an unconditional acceptance would require.

38.R RACI / Control Ownership

Table 48. Chapter 38 RACI / control ownership.

Function	Accountable (NCIE-008)	Responsible (Roadmap)	Consulted	Informed
Confirm acceptance/sign-off integration	Accountable Authority Class (to be assigned)	Acceptance Integration Owner (to be assigned)	NCIE-008/016 owners	Assurance/Audit (NCIE-008)

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR17-38-1: Confirm acceptance/sign-off integration.

Traceability: NCIE-001–016.

Human Review Register & Open Roadmap Decisions

Consolidate every unresolved roadmap decision exposed across Chapters 1-38 and Chapter 40 into one register: deduplicated, consistently classified, each stating precisely what it blocks. No item here manufactures closure of a decision the approved upstream chain, the approved NCIE-008–016 governance, security, persistence, domain-business, operations and verification layers, or the approved NCIE-017 Skeletal v0.1 does not itself settle. Blocking HR1–HR16 items are separately intaken as dependencies in Chapter 4 (Table 6) rather than cloned into this register.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-008–016 Human Review registers (referenced via Chapter 4 intake), plus roadmap-specific unresolved items from Chapters 1-38.

39.0 Salient Things to Note

Deduplicate upstream HR dependencies; create HR17 only for distinct roadmap/sequencing decisions.

39.1 Purpose

This register exists so that no unresolved roadmap decision is silently settled by expansion prose, and so that no chapter's own Human Review Focus item is left outside the register it is meant to feed — a defect found and corrected in the companion NCIE-008 v1.0 document's initial build. Every item from Chapters 1-38 below was extracted programmatically from that chapter's own Human Review Focus section (§39.4). Chapter 39's own item (HR17-39-1) and Chapter 40's own item (HR17-40-1) are each defined once as a fixed value and rendered identically both here and in their own chapter's Human Review Focus bullet, so neither can be omitted from the register or drift from the bullet that also states it.

39.2 Summary

Table 49. Human Review Register summary (mechanically computed).

Classification	Count
ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED	4
INSTITUTIONAL CONFIRMATION REQUIRED	30
RESOLVED BY UPSTREAM BASELINE	1
SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED	2
SOURCE/WORKFLOW DISCOVERY REQUIRED	3
Total	40
— Blocking	38
— Non-Blocking	2

39.3 Complete Register

Table 50. Complete Human Review Register.

Review ID	Source Chapter	Question	Classification	Blocking Scope	Upstream Cross-Reference	Status
HR17-1-1	Chapter 1	Confirm the NCIE-017 formal title reconciliation (§0.1 — corrected to	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm the NCIE-017 formal title reconciliation (§0.1 — corrected to	N/A — distinct roadmap	Open

			'Implementation Roadmap & Build Plan' against Master Controlled Register v2.0), and roadmap acceptance authority (skeletal Preliminary Human Review Theme 1).		'Implementation Roadmap & Build Plan' against Master Controlled Register v2.0), and roadmap acceptance authority (skeletal Preliminary Human Review Theme 1).	decision	
HR17-2-1	Chapter 2	Approve planning doctrine and roadmap ownership.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Approve planning doctrine and roadmap ownership.	N/A — distinct roadmap decision	Open	
HR17-3-1	Chapter 3	Confirm baseline-freeze authority/version register (Table 4).	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm baseline-freeze authority/version register (Table 4).	N/A — distinct roadmap decision	Open	
HR17-4-1	Chapter 4	Confirm treatment/ownership of Blocking upstream HR items (Table 6).	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm treatment/ownership of Blocking upstream HR items (Table 6).	N/A — distinct roadmap decision	Open	
HR17-5-1	Chapter 5	Confirm implementation traceability ownership.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm implementation traceability ownership.	N/A — distinct roadmap decision	Open	
HR17-6-1	Chapter 6	Approve WBS hierarchy/identifier convention (Table 9).	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Approve WBS hierarchy/identifier convention (Table 9).	N/A — distinct roadmap decision	Open	
HR17-7-1	Chapter 7	Confirm work-package ownership/evidence standard.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm work-package ownership/evidence standard.	N/A — distinct roadmap decision	Open	
HR17-8-1	Chapter 8	Confirm dependency classification/escalation (Table 12).	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm dependency classification/escalation (Table 12).	N/A — distinct roadmap decision	Open	
HR17-9-1	Chapter 9	Confirm sequencing/critical-path governance (Table 14; no calendar date assigned by this document).	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm sequencing/critical-path governance (Table 14; no calendar date assigned by this document).	N/A — distinct roadmap decision	Open	
HR17-10-1	Chapter 10	Confirm workstream structure and	INSTITUTIONAL CONFIRMATION REQUIRED	Confirm workstream	N/A — distinct	Open	

		coordination authority.	(Blocking)	structure and coordination authority.	roadmap decision	
HR17-11-1	Chapter 11	Confirm milestone/gate catalogue.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm milestone/gate catalogue.	N/A — distinct roadmap decision	Open
HR17-12-1	Chapter 12	Confirm entry/exit evidence and gate authority (Table 18).	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm entry/exit evidence and gate authority (Table 18).	N/A — distinct roadmap decision	Open
HR17-13-1	Chapter 13	Confirm implementation/Codex handover authority and package contents (Table 20).	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm implementation/Codex handover authority and package contents (Table 20).	N/A — distinct roadmap decision	Open
HR17-14-1	Chapter 14	Confirm prioritization authority/criteria.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm prioritization authority/criteria.	N/A — distinct roadmap decision	Open
HR17-15-1	Chapter 15	Confirm foundation technology decisions that block sequencing.	ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED (Blocking)	Confirm foundation technology decisions that block sequencing.	N/A — distinct roadmap decision	Open
HR17-16-1	Chapter 16	Confirm governance/security implementation prerequisites.	SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED (Blocking)	Confirm governance/security implementation prerequisites.	N/A — distinct roadmap decision	Open
HR17-17-1	Chapter 17	Confirm persistence technology/HR14 dependencies.	ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED (Blocking)	Confirm persistence technology/HR14 dependencies.	N/A — distinct roadmap decision	Open
HR17-18-1	Chapter 18	Confirm external/provider discovery and connector prerequisites.	SOURCE/WORKFLOW DISCOVERY REQUIRED (Blocking)	Confirm external/provider discovery and connector prerequisites.	N/A — distinct roadmap decision	Open
HR17-19-1	Chapter 19	Confirm AI/Agent implementation prerequisites.	SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED (Blocking)	Confirm AI/Agent implementation prerequisites.	N/A — distinct roadmap decision	Open
HR17-20-1	Chapter 20	Confirm Evidence/audit minimum enabling	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm Evidence/audit minimum enabling	N/A — distinct roadmap	Open

		sequence.		sequence.	decision	
HR17-21-1	Chapter 21	Confirm domain-rule/KPI dependencies.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm domain-rule/KPI dependencies.	N/A — distinct roadmap decision	Open
HR17-22-1	Chapter 22	Confirm UX implementation/prototype dependencies; does not block the state-presentation predecessor discipline already defined in the interim.	INSTITUTIONAL CONFIRMATION REQUIRED (Non-Blocking)	Non-blocking — does not gate chapter-level roadmap activity	N/A — distinct roadmap decision	Open
HR17-23-1	Chapter 23	Confirm infrastructure/platform prerequisites.	ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED (Blocking)	Confirm infrastructure/platform prerequisites.	N/A — distinct roadmap decision	Open
HR17-24-1	Chapter 24	Confirm NCIE-016 execution/acceptance sequencing.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm NCIE-016 execution/acceptance sequencing.	N/A — distinct roadmap decision	Open
HR17-25-1	Chapter 25	Confirm environment topology/configuration authority (no topology invented by this document).	ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED (Blocking)	Confirm environment topology/configuration authority (no topology invented by this document).	N/A — distinct roadmap decision	Open
HR17-26-1	Chapter 26	Confirm migration sources/cutover authority.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm migration sources/cutover authority.	N/A — distinct roadmap decision	Open
HR17-27-1	Chapter 27	Confirm external systems/provider sequencing.	SOURCE/WORKFLOW DISCOVERY REQUIRED (Blocking)	Confirm external systems/provider sequencing.	N/A — distinct roadmap decision	Open
HR17-28-1	Chapter 28	Confirm operational ownership/runbook readiness authority.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm operational ownership/runbook readiness authority.	N/A — distinct roadmap decision	Open
HR17-29-1	Chapter 29	Confirm release/cutover/go-live authority.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm release/cutover/go-live authority.	N/A — distinct roadmap decision	Open
HR17-30-1	Chapter 30	Confirm rollback/recovery	INSTITUTIONAL CONFIRMATION REQUIRED	Confirm rollback/recovery	N/A — distinct roadmap	Open

		decision authority.	(Blocking)	decision authority.	p decision	
HR17-31-1	Chapter 31	Confirm skill/resource availability assumptions (Table 39; no headcount/budget invented by this document).	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm skill/resource availability assumptions (Table 39; no headcount/budget invented by this document).	N/A — distinct roadmap decision	Open
HR17-32-1	Chapter 32	Confirm estimate/schedule authority and planning horizon (no date/duration invented by this document).	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm estimate/schedule authority and planning horizon (no date/duration invented by this document).	N/A — distinct roadmap decision	Open
HR17-33-1	Chapter 33	Confirm procurement/provider dependencies (no lead time/contract invented by this document).	SOURCE/WORKFLOW DISCOVERY REQUIRED (Blocking)	Confirm procurement/provider dependencies (no lead time/contract invented by this document).	N/A — distinct roadmap decision	Open
HR17-34-1	Chapter 34	Confirm RAID ownership/risk-acceptance authority (Table 43).	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm RAID ownership/risk-acceptance authority (Table 43).	N/A — distinct roadmap decision	Open
HR17-35-1	Chapter 35	Confirm re-baselining/change authority.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm re-baselining/change authority.	N/A — distinct roadmap decision	Open
HR17-36-1	Chapter 36	Confirm progress-reporting evidence/status rules.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm progress-reporting evidence/status rules.	N/A — distinct roadmap decision	Open
HR17-37-1	Chapter 37	Confirm Definition of Ready/Done.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm Definition of Ready/Done.	N/A — distinct roadmap decision	Open
HR17-38-1	Chapter 38	Confirm acceptance/sign-off integration.	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Confirm acceptance/sign-off integration.	N/A — distinct roadmap decision	Open
HR17-39-1	Chapter 39	Review this consolidated register and decide, for each Blocking item, whether it must be	INSTITUTIONAL CONFIRMATION REQUIRED (Blocking)	Review this consolidated register and decide, for each Blocking item, whether it	N/A — distinct roadmap decision	Open

		resolved before NCIE-017 v1.0 may be approved or may instead remain open under an interim design default.		must be resolved before NCIE-017 v1.0 may be approved or may instead remain open under an interim design default.		
HR17-40-1	Chapter 40	Approve the NCIE-017 skeletal structure and authorize controlled full expansion — already satisfied by the approved NCIE-017 Skeletal v0.1 and the production directive that authorized this v1.0 expansion; preserved here as historical record, not a live open item.	RESOLVED BY UPSTREAM BASELINE (Non-Blocking)	Non-blocking — does not gate chapter-level roadmap activity	N/A — distinct roadmap decision	Resolved by upstream baseline

39.4 Mechanical Verification

38 of these 40 items were recomputed programmatically from the Human Review Focus bullet text of Chapters 1-38 rather than assembled by hand. The remaining 2 items are HR17-39-1 (this chapter's own item) and HR17-40-1 (Chapter 40's own item), each defined once and rendered identically in both this table and their own chapter's Human Review Focus bullet. 0 duplicate Review IDs; 0 orphan Human Review items — every Human Review ID appearing anywhere in this specification is reconciled into this register.

39.5 Upstream Deduplication (Directive Item 73)

Every NCIE-017 Human Review item was individually checked against the existing HR1–HR16 registers. No NCIE-017 item was found to restate an identical question verbatim — each concerns a distinct roadmap/sequencing decision (WBS/dependency structure, workstream ownership, estimate/schedule authority, RAID ownership) that those upstream documents raise the need for but do not themselves resolve. Blocking HR1–HR16 items are handled as explicit dependency intake in Chapter 4 (Table 6), not cloned into HR17, per directive item 73's instruction not to populate this register with upstream questions merely to fill it.

39.6 Blocking Scope Discipline

Each Blocking item states exactly what it blocks — a specific WBS decision, dependency classification, workstream prerequisite, estimate/schedule authority or acceptance-gate integration — rather than implying the whole of NCIE-017 or the whole approved upstream chain is blocked. Non-Blocking items are explicitly proposed defaults or confirmations that do not gate the corresponding chapter's roadmap activity in the interim.

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] (Blocking) HR17-39-1: Review this consolidated register and decide, for each Blocking item, whether it must be resolved before NCIE-017 v1.0 may be approved or may instead remain open under an interim design default.

Traceability: All chapters above (mechanically reconciled register); NCIE-008–016 Human Review registers (via Chapter 4 intake).

Traceability, Acceptance & Final Implementation Handover

Close traceability to the approved NCIE-001-016 foundation and the approved NCIE-017 Skeletal v0.1; define roadmap acceptance criteria and the final controlled handover to implementation execution and change governance.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW. Planning specification only; nothing in this chapter implies implementation work has started, tests have passed, blockers have cleared, or release/go-live is authorized.

Upstream: Approved NCIE-001-016 foundation.

40.0 Salient Things to Note

Skeleton approval authorizes expansion only; roadmap approval is not execution or go-live authorization — §40.4's acceptance gate states this explicitly, and no implementation work described in this specification has started.

40.1 Traceability Matrix

Every chapter of this document traces to an approved upstream authority or to the approved NCIE-017 Skeletal v0.1 itself. No chapter states a sequencing requirement without an identifiable source. As the final document in the NCIE-001–017 suite, this traceability matrix closes the authority chain that began with NCIE-001.

Table 51. Full traceability matrix.

NCIE-017 Chapter	Primary Upstream Authority	Relationship
Chapter 1	NCIE-002 Ch.32; NCIE-014 Ch.34; approved NCIE-001-016 chain	Mandate/authority chain/roadmap boundary
Chapter 2	NCIE-001-016	Implementation planning doctrine
Chapter 3	NCIE-001-016	Approved baseline & architecture freeze register
Chapter 4	NCIE-001-016	Human Review dependency intake
Chapter 5	NCIE-001-016	Requirement-to-implementation traceability
Chapter 6	NCIE-001-016	WBS architecture
Chapter 7	NCIE-001-016	Work package definition standard
Chapter 8	NCIE-001-016	Dependency graph & predecessor logic
Chapter 9	NCIE-001-016	Build sequencing & critical-path method
Chapter 10	NCIE-001-016	Parallelization & workstream coordination
Chapter 11	NCIE-001-016	Milestone & stage-gate architecture
Chapter 12	NCIE-001-016	Entry/exit criteria by stage
Chapter 13	NCIE-002 Ch.32 architecture-to-Codex handover	Architecture-to-Codex/implementation handover
Chapter 14	NCIE-001-016	Implementation backlog & prioritization
Chapter 15	NCIE-001-016	Foundation/platform workstream
Chapter 16	NCIE-001-016	Security, IAM & governance workstream
Chapter 17	NCIE-001-016	Data, database & storage

		workstream
Chapter 18	NCIE-001-016	API, integration & connector workstream
Chapter 19	NCIE-001-016	ARGUS, Memory, Agent & AI workstream
Chapter 20	NCIE-001-016	Evidence, provenance & audit workstream
Chapter 21	NCIE-001-016	Functional Modules workstream
Chapter 22	NCIE-001-016	UX/UI & Design System workstream
Chapter 23	NCIE-001-016	DevSecOps, infrastructure & deployment workstream
Chapter 24	NCIE-016 approved verification architecture	Testing, verification & acceptance workstream
Chapter 25	NCIE-001-016	Environment & configuration sequencing
Chapter 26	NCIE-001-016	Data migration & initial-load sequencing
Chapter 27	NCIE-001-016	External systems/provider integration sequencing
Chapter 28	NCIE-001-016	Operational readiness & runbook sequencing
Chapter 29	NCIE-001-016	Release, cutover & go-live dependency plan
Chapter 30	NCIE-001-016	Rollback, recovery & contingency planning
Chapter 31	NCIE-001-016	Resource, skill & ownership planning
Chapter 32	NCIE-001-016	Estimate, duration & schedule governance
Chapter 33	NCIE-001-016	Procurement/external dependency tracking
Chapter 34	NCIE-001-016	RAID register
Chapter 35	NCIE-001-016	Change control & roadmap re-baselining
Chapter 36	NCIE-001-016	Implementation evidence & progress reporting
Chapter 37	NCIE-001-016	Definition of Ready/Done
Chapter 38	NCIE-001-016	Acceptance-gate & sign-off integration
Chapter 39	All chapters above; NCIE-008-016 Human Review registers	Consolidated Human Review Register
Chapter 40	Approved NCIE-001-016; NCIE-017 Skeletal v0.1	Traceability/acceptance/final handover

40.2 Chapter-Count and Salient-Note Mechanical Verification

A programmatic scan of this document confirms exactly 40 sequential 'NCIE-017 — Chapter N Expansion' labels, N = 1..40, each present exactly once, in order, with no gap, duplicate or unintended renaming (directive item 84). A second scan confirms an explicit '<N>.0 Salient Things to Note' heading present for all 40 chapters (directive item 85), rerun independently against this final document rather than assumed from the build script.

40.3 Skeleton-to-Full Fidelity Verification

Every approved Chapter Purpose, Salient Things to Note, Principal Component, Upstream Dependency/Authority and Human Review Focus from the approved NCIE-017 Skeletal v0.1 is substantively represented in the corresponding full chapter above (directive item 83); none is omitted, and none is conceptually inverted.

40.4 Acceptance Criteria and the Unresolved-Item Gate

Full approval of NCIE-017 v1.0 requires: (a) 40/40 chapter integrity confirmed; (b) 40/40 explicit Salient Things to Note subsections confirmed; (c) skeleton-to-full fidelity confirmed; (d) every Blocking Human Review item in Chapter 39, and every intaken HR1-16 dependency in Chapter 4, either resolved by an institutional decision or explicitly and knowingly deferred by the approving authority (including the formal-title reconciliation, §0.1, HR17-1-1); (e) confirmation that no date, budget, resource, platform selection, test result, or acceptance decision anywhere in this document is falsely presented as fixed, executed, or granted; and (f) three-artifact consistency (this document, the Change Register and the Verification Report) confirmed. Approval of the v0.1 skeleton authorized this expansion only — it did not itself approve any date, budget, resource assignment, platform selection or release authority proposed above, and it did not authorize a single unit of implementation work to begin. Roadmap approval under this chapter authorizes a controlled planning baseline only; it does not constitute execution authorization, does not prove requirement correctness, implementation correctness, security assurance, recovery capability, performance capacity, or production readiness — all of which remain downstream of this document, resting on the already-approved NCIE-001–016 architecture and the actual, evidence-backed execution and verification this roadmap only sequences.

40.5 Final Implementation Handover Register

NCIE-017 does not preempt actual implementation execution or institutional acceptance. The table below states what is handed over and to whom, closing the full NCIE-001–017 documentation suite.

Table 52. Final implementation handover register.

Downstream Consumer	What NCIE-017 Hands Over
Implementation execution teams	The Master WBS (Table 9), dependency register (Table 12), roadmap sequence matrix (Table 14) and architecture-to-implementation handover matrix (Table 20) as the controlled planning baseline for actual build work; this document supplies the sequence, not the executed build.
NCIE-008 institutional acceptance authority	The consolidated Human Review Register (Chapter 39), HR1-16 dependency intake (Table 6) and acceptance-gate integration (Chapter 38) as the basis for eventual, evidence-backed go-live decisions once implementation and NCIE-016 verification are actually executed.
Change governance	The approved-baseline freeze register (Table 4) and change-control process (Chapter 35) as the mechanism by which any future architecture change re-enters upstream governance rather than being silently absorbed into a resequenced roadmap.

40.6 Readiness and Final Disposition

NCIE-017 Implementation Roadmap & Build Plan v1.0 — Full Expansion of Approved Skeletal v0.1 — is READY FOR HUMAN REVIEW. It is NOT marked Approved. The approved v0.1 skeleton remains historically preserved as the expansion architecture; every proposed date, budget, resource assignment, platform selection and release authority deferred in Chapters 1-38 remains a proposal pending the Chapter 39 Human Review Register and the Chapter 4 HR1-16 dependency intake. No implementation work described anywhere in this specification has started, no test referenced from NCIE-016 has been executed, and no release/go-live has been authorized — this document is a controlled planning baseline, not an implementation record. This is the final document in the NCIE-001-017 controlled documentation suite; its approval, together with the already-approved NCIE-001-016 foundation, would complete the suite's planning-stage coverage while leaving all actual execution, testing and acceptance downstream and unstarted.

Human Review Focus

- [RESOLVED BY UPSTREAM BASELINE] (Non-Blocking) HR17-40-1: Approve the NCIE-017 skeletal structure and authorize controlled full expansion — already satisfied by the approved NCIE-017 Skeletal v0.1 and the production directive that authorized this v1.0 expansion; preserved here as historical record, not a live open item.

Traceability: NCIE-001-016 approved foundation; approved NCIE-017 Skeletal v0.1.